

# Casual Document for Probabilistic Principal Component Analysis MLE Asymptotic Normality

Wen, Zehai

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# Chapter 1

## Low Dimensional Asymptotic Normality

### 1.1 Basic Setup

Given a "high dimensional" data set  $\{\mathbf{x}_i\}_{i=1}^n$  where  $\mathbf{x}_i \in \mathbb{R}^p$  for all  $i = 1, \dots, n$ , the statistician fixes a positive integer  $q < p$  and wishes to find the  $q$  principal components. Assuming that the data is centred (that is, with mean 0), the probabilistic principal component analysis model is a statistical model that treats  $\mathbf{x}_i$ 's as realizations of independent random variables  $\mathbf{X}_i = \mathbf{W}\mathbf{Z}_i + \boldsymbol{\epsilon}_i$ , where  $\mathbf{W}$  is an unknown  $p \times q$  loading matrix of full rank,  $\mathbf{Z}_i \sim N_q(\mathbf{0}, \mathbf{I}_q)$  is the latent normal variable, and  $\boldsymbol{\epsilon}_i \sim N_p(\mathbf{0}, \sigma^2 \mathbf{I}_p)$  is the Gaussian noise with unknown standard deviation  $\sigma > 0$ , and  $(\mathbf{Z}_1, \dots, \mathbf{Z}_n, \boldsymbol{\epsilon}_1, \dots, \boldsymbol{\epsilon}_n)$  are all independent (for now). Put  $\theta = (\mathbf{W}_\theta, \sigma_\theta^2)$  for the unknown parameter to estimate and  $\Theta$  for all possible choices of  $\theta$ .  $\Theta$  is thus the collection of all tuples  $(\mathbf{W}, \sigma^2)$  such that  $\mathbf{W}$  is a real  $p \times q$  matrix with  $\text{rank}(\mathbf{W}) = q$  and  $\sigma^2 \in (0, \infty)$ . Given that the collection of all real  $p \times q$  matrix with rank  $q$  is an open subset of the Euclidean space  $\mathbb{R}^{pq}$ , we consider  $\Theta$  as an open subset of the Euclidean space  $\mathbb{R}^{pq+1}$  and our statistical model is

$$\mathcal{P} = \{N_p(\mathbf{0}, \mathbf{W}_\theta \mathbf{W}_\theta^T + \sigma_\theta^2 \mathbf{I}_p)\}_{\theta \in \Theta}. \quad (1.1)$$

To resolve the non-identifiability of  $\mathcal{P}$ , we introduce the quotient manifold structure. For  $(\mathbf{A}, s) \in \Theta$  and  $(\mathbf{B}, t) \in \Theta$ , we define  $(\mathbf{A}, s) \sim (\mathbf{B}, t)$  to mean that they give rise to the same covariance matrix, namely when  $\mathbf{A}\mathbf{A}^T + s\mathbf{I}_p = \mathbf{B}\mathbf{B}^T + t\mathbf{I}_p$ . The equivalence classes are denoted by  $[(\mathbf{A}, s)]$  or more briefly  $[\mathbf{A}, s]$ . Unless otherwise mentioned,  $(\mathbf{A}, s) \in \Theta$  is the default representative we pick when we write an equivalence class as  $[\mathbf{A}, s]$ . Writing  $O(q)$  for the collection of all  $q \times q$  real orthogonal matrices, it was shown in Datta's thesis that  $[\mathbf{A}, s] = \{(\mathbf{A}\mathbf{Q}, s) \mid \mathbf{Q} \in O(q)\}$ . Therefore,  $M := \Theta / \sim$  can be identified as the orbits of the right action of  $O(q)$  on  $\Theta$  and, given that this Lie group action is smooth, proper, and free, the quotient manifold theorem (see [Lee, 2013](#), Theorem 21.10) shows that there is a unique smooth structure making  $M$  a manifold of dimension

$$d = pq + 1 - \frac{q(q-1)}{2} \quad (1.2)$$

and the map  $\pi : \Theta \rightarrow M$  defined by

$$\pi((\mathbf{A}, s)) = [\mathbf{A}, s] \quad (1.3)$$

is a smooth submersion. Moreover, this action is vertical, transitive on fibers and is isometric with respect to the Euclidean metric. Thus, there exists a unique metric  $g$  on  $M$  making  $(M, g)$  a Riemannian manifold and  $\pi$  a Riemannian submersion (see [Lee, 2018](#), Theorem 2.28). In more concrete terms, if  $g^\Theta$  is the standard Euclidean metric given by

$$g_{(\mathbf{A}, s)}^\Theta((\mathbf{V}, c), (\mathbf{U}, b)) = \text{tr}(\mathbf{V}\mathbf{U}^T) + cb, \quad (1.4)$$

whenever  $(\mathbf{A}, s) \in \Theta$ ,  $(\mathbf{V}, c), (\mathbf{U}, b) \in T_{(\mathbf{A}, s)}\Theta \cong \mathbb{R}^{p \times q} \times \mathbb{R}$ , then

$$g_{[\mathbf{A}, s]}(\mathbf{v}, \mathbf{h}) = g_{[\mathbf{A}, s]}(d\pi_{(\mathbf{A}, s)}((\mathbf{V}, c)), d\pi_{(\mathbf{A}, s)}((\mathbf{U}, b))) = g_{(\mathbf{A}, s)}^\Theta((\mathbf{V}, c), (\mathbf{U}, b)), \quad (1.5)$$

whenever  $\mathbf{v}, \mathbf{h} \in T_{[\mathbf{A}, s]}M$  lifting to  $(\mathbf{V}, c)$  and  $(\mathbf{U}, b)$  respectively. A standard fact is that  $\Theta$  is connected. Consequently,  $M$  is also connected. For later convenience, let us define two functions  $\Phi : \Theta \rightarrow \mathbb{R}^{p \times p}$  and  $\Sigma : M \rightarrow \mathbb{R}^{p \times p}$  by setting, for every  $[\mathbf{A}, s] \in M$ :

$$\Sigma([\mathbf{A}, s]) := \Phi([\mathbf{A}, s]) := \mathbf{A}\mathbf{A}^T + s\mathbf{I}_p. \quad (1.6)$$

In other words,  $\Sigma \circ \pi = \Phi$  gives the covariance matrix. (1.1) is then reparametrized as

$$\mathcal{P} = \{N_p(\mathbf{0}, \Sigma([\mathbf{A}, s]))\}_{[\mathbf{A}, s] \in M} \quad (1.7)$$

and this is the identifiable model we shall work with.

Finally, throughout the entire document, our default norm for a matrix  $\mathbf{A}$  will be the spectral norm, denoted by  $\|\mathbf{A}\|$ . We will also make use of the Frobenius norm  $\|\mathbf{A}\|_F$  as well. The eigenvalues and the singular values of a matrix will always be labelled in decreasing order. In symbols, if  $\mathbf{A}$  is a matrix of size  $p \times q$ , then we denote its  $q$  singular values as

$$\|\mathbf{A}\| = \sigma_1(\mathbf{A}) \geq \sigma_2(\mathbf{A}) \geq \cdots \geq \sigma_q(\mathbf{A}) = \sigma_{\min}(\mathbf{A}) \geq 0$$

and, if  $\mathbf{S}$  is a symmetric matrix of size  $p \times p$ , then we denote its  $p$  eigenvalues as

$$\lambda_1(\mathbf{S}) \geq \lambda_2(\mathbf{S}) \geq \cdots \geq \lambda_p(\mathbf{S}).$$

## 1.2 Geometric Facts about $(M, g)$

### 1.2.1 Horizontal and Verticle Tangent Spaces

Working with Riemannian submersion, we shall as usual start with computations regarding the horizontal and vertical tangent spaces.

#### Proposition 1.

1. The map  $\Sigma$  defined in (1.6) is a smooth injective immersion.
2. At every  $(\mathbf{A}, s) \in \Theta$ , the vertical tangent space

$$V_{(\mathbf{A}, s)} := \ker(d\pi_{(\mathbf{A}, s)}) = \ker(d\Phi_{(\mathbf{A}, s)}) \quad (1.8)$$

equals to the collection of all tangent vectors of the form  $(\mathbf{A}\mathbf{\Omega}, 0) \in T_{(\mathbf{A}, s)}\Theta$ , with  $\mathbf{\Omega}$  ranged over all  $q \times q$  skew symmetric matrices.

3. At every  $(\mathbf{A}, s) \in \Theta$ , the horizontal tangent space  $H_{(\mathbf{A}, s)} := V_{(\mathbf{A}, s)}^\perp$  equals to the collection of all tangent vectors of the form  $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)}\Theta$  ranged over all  $c \in \mathbb{R}$  and all  $p \times q$  matrices  $\mathbf{V}$  satisfying  $\mathbf{V}^T \mathbf{A} = \mathbf{A}^T \mathbf{V}$ .

*Proof.* Fix a point  $(\mathbf{A}, s) \in \Theta$  throughout this proof. Before we start the proof, let us first study the set  $\ker(d\Phi_{(\mathbf{A}, s)})$ , where  $\Phi$  is given in (1.6). For  $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)}\Theta$ , define a curve  $\gamma : (-\epsilon, \epsilon) \rightarrow \Theta$  by setting  $\gamma(t) = (\mathbf{A} + t\mathbf{V}, s + tc)$ . Then:

$$d\Phi_{(\mathbf{A}, s)}((\mathbf{V}, c)) = \left. \frac{d}{dt} \right|_{t=0} \Phi(\gamma(t)) = \mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + c\mathbf{I}_p.$$

If  $(\mathbf{V}, c) \in \ker(d\Phi_{(\mathbf{A}, s)})$ , then

$$\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + c\mathbf{I}_p = \mathbf{0}. \quad (1.9)$$

If  $\mathbf{u} \in \mathbb{R}^p$  is any nonzero vector orthogonal to all column vectors of  $\mathbf{A}$ , then  $\mathbf{A}^T \mathbf{u} = \mathbf{0}$  and

$$0 = \mathbf{u}^T (\mathbf{A} \mathbf{V}^T + \mathbf{V} \mathbf{A}^T + c \mathbf{I}_p) \mathbf{u} = c \mathbf{u}^T \mathbf{u}.$$

Given that  $\mathbf{u} \neq \mathbf{0}$ , we must have  $c = 0$  and (1.9) reduces to

$$\mathbf{A} \mathbf{V}^T + \mathbf{V} \mathbf{A}^T = \mathbf{0}. \quad (1.10)$$

Apply  $\mathbf{u}$  to the right to get:

$$\mathbf{A} \mathbf{V}^T \mathbf{u} = \mathbf{0}.$$

Given that  $\mathbf{A}$  has full rank, rank nullity theorem forces

$$\mathbf{V}^T \mathbf{u} = \mathbf{0}.$$

Therefore, we have proven that every vector  $\mathbf{u}$  orthogonal to columns vectors of  $\mathbf{A}$  will also be orthogonal to column vectors of  $\mathbf{V}$ . This means  $\mathbf{V} = \mathbf{A} \mathbf{\Omega}$  for some  $q \times q$  matrix  $\mathbf{\Omega}$ . Substitute this back to (1.10) gives

$$\mathbf{A}(\mathbf{\Omega}^T + \mathbf{\Omega})\mathbf{A}^T = \mathbf{0}.$$

Applying  $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$  to the left and  $\mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1}$  to the right yields  $\mathbf{\Omega}^T + \mathbf{\Omega} = \mathbf{0}$ . Therefore,  $\mathbf{\Omega}$  is skew-symmetric. Conversely, (1.10) is satisfied by  $\mathbf{V} = \mathbf{A} \mathbf{\Omega}$  with any skew-symmetric matrix  $\mathbf{\Omega}$ . We have proven that  $\ker(d\Phi_{(\mathbf{A}, s)})$  equals to the collection of all tangent vectors of the form  $(\mathbf{A} \mathbf{\Omega}, 0) \in T_{(\mathbf{A}, s)} \Theta$ , with  $\mathbf{\Omega}$  ranged over all  $q \times q$  skew symmetric matrices.

We now move back to prove 1 and 2 at the same time by showing (1.8). Clearly,  $\Sigma$  is well-defined and is injective. The smoothness of  $\Sigma$  is guaranteed by relation  $\Sigma \circ \pi = \Phi$  (see for example Lee, 2013, Theorem 4.29). By the chain rule, we have

$$d\Phi_{(\mathbf{A}, s)} = d\Sigma_{[\mathbf{A}, s]} \circ d\pi_{(\mathbf{A}, s)}. \quad (1.11)$$

It follows immediately that  $\ker(d\pi_{(\mathbf{A}, s)}) \subseteq \ker(d\Phi_{(\mathbf{A}, s)})$ . The other direction will follow if we can prove that  $d\Sigma_{[\mathbf{A}, s]}$  is injective.

Suppose that  $\mathbf{v} \in T_{[\mathbf{A}, s]} M$  is such that  $d\Sigma_{[\mathbf{A}, s]}(\mathbf{v}) = \mathbf{0}$ . We prove that  $\mathbf{v} = \mathbf{0}$ . Let  $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)} \Theta$  be the horizontal lift of  $\mathbf{v}$  at  $(\mathbf{A}, s)$ , meaning that  $(\mathbf{V}, c)$  is the unique horizontal tangent vector at  $(\mathbf{A}, s)$  such that  $\mathbf{v} = d\pi_{(\mathbf{A}, s)}((\mathbf{V}, c))$ . (1.11) then implies that  $(\mathbf{V}, c) \in \ker(d\Phi_{(\mathbf{A}, s)})$  and the preceding discussion shows that  $c = 0$  and  $\mathbf{V} = \mathbf{A} \mathbf{\Omega}$  for some skew-symmetric  $\mathbf{\Omega}$ . To proceed, we look at the curve  $\gamma : (-\epsilon, \epsilon) \rightarrow \Theta$  defined by  $\gamma(t) = (\mathbf{A} e^{t\mathbf{\Omega}}, s)$ . Clearly  $\gamma(0) = (\mathbf{A}, s)$  and  $\gamma'(0) = (\mathbf{A} \mathbf{\Omega}, 0)$ . Moreover, the matrix  $e^{t\mathbf{\Omega}}$  is always orthogonal for any  $t$  and any skew-symmetric  $\mathbf{\Omega}$  so that  $\pi(\gamma(t)) = [\mathbf{A}, s]$  for all  $t$ . We have

$$\mathbf{v} = d\pi_{(\mathbf{A}, s)}((\mathbf{A} \mathbf{\Omega}, 0)) = d\pi_{(\mathbf{A}, s)}(\gamma'(0)) = \left. \frac{d}{dt} \right|_{t=0} \pi(\gamma(t)) = \mathbf{0} \in T_{[\mathbf{A}, s]} M.$$

Therefore,  $d\Sigma_{[\mathbf{A}, s]}$  is injective and the vertical tangent space is the same as  $\ker(d\Phi_{(\mathbf{A}, s)})$ . Since  $[\mathbf{A}, s]$  is arbitrary,  $\Sigma$  is an immersion.

To compute the horizontal tangent space, any  $(\mathbf{V}, c) \in V_{(\mathbf{A}, s)}^\perp$  will satisfy

$$0 = g_{(\mathbf{A}, s)}^\Theta((\mathbf{V}, c), (\mathbf{A} \mathbf{\Omega}, 0)) = \text{tr}(\mathbf{A}^T \mathbf{V} \mathbf{\Omega}) \quad (1.12)$$

for all skew-symmetric  $\mathbf{\Omega}$ . Decomposing  $\mathbf{A}^T \mathbf{V} = \mathbf{S} + \mathbf{K}$ , where  $\mathbf{S}$  is the symmetric part and  $\mathbf{K}$  is the skew-symmetric part, we find out that

$$0 = \text{tr}(\mathbf{S} \mathbf{\Omega}) + \text{tr}(\mathbf{K} \mathbf{\Omega}) = 0 + \text{tr}(\mathbf{K} \mathbf{\Omega}) = \text{tr}(\mathbf{K} \mathbf{\Omega}).$$

Setting  $\mathbf{\Omega} = \mathbf{K}^T$  will yield that  $\mathbf{K} = \mathbf{0}$ . Therefore,  $\mathbf{A}^T \mathbf{V}$  is a symmetric matrix. Conversely, any  $\mathbf{V}$  such that  $\mathbf{A}^T \mathbf{V}$  is symmetric will satisfy (1.12). The proof is complete.  $\square$

We will need to compute the projection of a tangent vector to horizontal and verticle subspaces.

**Lemma 2.** For any  $q \times q$  symmetric invertible matrix  $\mathbf{S}$  and  $q \times q$  matrix  $\mathbf{\Omega}$ , define

$$\psi_{\mathbf{S}}(\mathbf{\Omega}) = \mathbf{S}\mathbf{\Omega} + \mathbf{\Omega}\mathbf{S}.$$

Then, for each fixed  $\mathbf{S}$ ,  $\psi_{\mathbf{S}}$  is linear, maps skew-symmetric matrices to skew-symmetric matrices, and is invertible. Furthermore, both  $\psi_{\mathbf{S}}(\mathbf{\Omega})$  and  $\psi_{\mathbf{S}}^{-1}(\mathbf{\Omega})$  depend smoothly on the tuple  $(\mathbf{S}, \mathbf{\Omega})$ .

*Proof.* It is clear from definition that  $\psi_{\mathbf{S}}$  is linear and  $\psi_{\mathbf{S}}(\mathbf{\Omega})$  is skew-symmetric whenever  $\mathbf{\Omega}$  is. To show that  $\psi_{\mathbf{S}}$  is invertible, suppose that  $\mathbf{\Omega}$  is such that

$$\psi_{\mathbf{S}}(\mathbf{\Omega}) = \mathbf{0}.$$

Apply spectral theorem to get  $\mathbf{Q} \in O(q)$  and a  $q \times q$  diagonal matrix  $\Lambda$  with positive entries  $\lambda_1(\mathbf{S}), \dots, \lambda_q(\mathbf{S})$  such that  $\mathbf{S} = \mathbf{Q}\Lambda\mathbf{Q}^T$ . Substitution yields

$$\Lambda\mathbf{Q}^T\mathbf{\Omega}\mathbf{Q} + \mathbf{Q}^T\mathbf{\Omega}\mathbf{Q}\Lambda = \mathbf{0}.$$

Looking at each entry, we arrive at

$$(\lambda_i(\mathbf{S}) + \lambda_j(\mathbf{S}))(\mathbf{Q}^T\mathbf{\Omega}\mathbf{Q})_{ij} = 0$$

for  $i, j = 1, \dots, q$ . Given that  $\mathbf{S}$  is invertible, the eigenvalues of  $\mathbf{S}$  are positive. Thus,  $\mathbf{Q}^T\mathbf{\Omega}\mathbf{Q} = \mathbf{0}$  and  $\mathbf{\Omega} = \mathbf{0}$ . We have proved that  $\psi_{\mathbf{S}}$  is invertible. Given that  $\psi_{\mathbf{S}}(\mathbf{\Omega})$  is a polynomial of the tuple  $(\mathbf{S}, \mathbf{\Omega})$ , it depends smoothly on the tuple.  $\psi_{\mathbf{S}}^{-1}(\mathbf{\Omega})$  also depends smoothly on  $(\mathbf{S}, \mathbf{\Omega})$  because the inverse is a smooth operation.  $\square$

**Proposition 3.** Given  $(\mathbf{A}, s) \in \Theta$  and  $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)}\Theta$ , orthogonal projection of  $(\mathbf{V}, c)$  to  $V_{(\mathbf{A}, s)}$  is

$$(\mathbf{A}\psi_{\mathbf{A}^T\mathbf{A}}^{-1}(\mathbf{A}^T\mathbf{V} - \mathbf{V}^T\mathbf{A}), 0)$$

and the same vector projected to  $H_{(\mathbf{A}, s)}$  is

$$(\mathbf{V} - \mathbf{A}\psi_{\mathbf{A}^T\mathbf{A}}^{-1}(\mathbf{A}^T\mathbf{V} - \mathbf{V}^T\mathbf{A}), c). \quad (1.13)$$

Here  $\psi$  is as in Lemma 2.

*Proof.* It suffices to prove the formula for verticle projection as the horizontal one follows by a subtraction. By Proposition 1, the orthogonal projection of  $(\mathbf{V}, c)$  to  $V_{(\mathbf{A}, s)}$  equals to  $(\mathbf{A}\mathbf{\Omega}, 0) \in V_{(\mathbf{A}, s)}$  for some  $q \times q$  skew-symmetric matrix  $\mathbf{\Omega}$  which minimizes  $\|\mathbf{V} - \mathbf{A}\mathbf{\Omega}\|_F^2$ . To compute  $\mathbf{\Omega}$ , let us compute the directional derivative of the quantity in the direction of a  $q \times q$  skew-symmetric matrix  $\mathbf{H}$ . For any  $t \in \mathbb{R}$ :

$$\begin{aligned} \|\mathbf{V} - \mathbf{A}(\mathbf{\Omega} + t\mathbf{H})\|_F^2 &= \text{tr} \left\{ [\mathbf{V} - \mathbf{A}(\mathbf{\Omega} + t\mathbf{H})][\mathbf{V} - \mathbf{A}(\mathbf{\Omega} + t\mathbf{H})]^T \right\} \\ &= \|\mathbf{V}\|_F^2 - 2\text{tr}(\mathbf{\Omega}^T\mathbf{A}^T\mathbf{V}) - 2t\text{tr}(\mathbf{H}^T\mathbf{A}^T\mathbf{V}) + \text{tr}(\mathbf{\Omega}^T\mathbf{A}^T\mathbf{A}\mathbf{\Omega}) + t^2\text{tr}(\mathbf{H}^T\mathbf{A}^T\mathbf{A}\mathbf{H}) \end{aligned}$$

and

$$\left. \frac{d}{dt} \right|_{t=0} \|\mathbf{V} - \mathbf{A}(\mathbf{\Omega} + t\mathbf{H})\|_F^2 = \text{tr} [\mathbf{H}^T(\mathbf{A}^T\mathbf{A}\mathbf{\Omega} - \mathbf{A}^T\mathbf{V})].$$

Therefore, any solution to the equation setting all directional derivatives to 0 yields that  $\text{tr} [\mathbf{H}^T(\mathbf{A}^T\mathbf{A}\mathbf{\Omega} - \mathbf{A}^T\mathbf{V})] = 0$  for all skew-symmetric  $\mathbf{H}$ , or equivalently that  $\mathbf{A}^T\mathbf{A}\mathbf{\Omega} - \mathbf{A}^T\mathbf{V}$  is symmetric. Therefore,  $\mathbf{\Omega}$  must satisfy

$$\mathbf{A}^T\mathbf{A}\mathbf{\Omega} + \mathbf{\Omega}\mathbf{A}^T\mathbf{A} = \mathbf{A}^T\mathbf{V} - \mathbf{V}^T\mathbf{A}.$$

The proof is complete once we apply Lemma 2.  $\square$

In most textbooks, one can find the standard fact that any smooth vector field in  $M$  uniquely horizontally lifts to a smooth vector field in  $\Theta$ . However, not all textbooks mention about horizontal lifting curves or vector fields along curves. We bookkeep such a proposition for later convenience.

**Proposition 4.** Given a smooth curve  $\gamma : (a, b) \rightarrow M$ , a point  $t_0 \in (a, b)$ , and a representative  $(\mathbf{A}, s)$  of  $\gamma(t_0)$ , there exists a unique smooth curve  $c : (a, b) \rightarrow \Theta$ , which we refer to as the horizontal lift of  $\gamma$  at  $t_0$  and representative  $(\mathbf{A}, s)$ , such that  $c(t_0) = (\mathbf{A}, s)$  and, for all  $t \in (a, b)$ ,  $\pi(c(t)) = \gamma(t)$ ,  $c'(t) \in H_{c(t)}$ , and  $d\pi_{c(t)}(c'(t)) = \gamma'(t)$ . Furthermore, given a smooth vector field  $\mathbf{X}(t)$  along  $\gamma$ , define, for each  $t \in (a, b)$ ,  $\mathbf{Y}(t)$  to be the horizontal lift of  $\mathbf{X}(t)$  at  $c(t)$ . Then  $\mathbf{Y}(t)$  is a smooth vector field along  $c$ .

*Proof.* The fact that  $\gamma$  can be uniquely horizontally lifted to a curve  $c$  is immediate from Theorem 17.8 and 17.9 from [Michor, 2008](#). Given a smooth vector field  $\mathbf{X}(t)$  along  $\gamma$  and any  $t \in (a, b)$ , we can choose a smooth local frame around  $\gamma(t)$  so that the coefficients of  $\mathbf{X}(t)$  under this frame is smooth. Given that smooth local frames in  $M$  lift horizontally to smooth local frames in  $\Theta$ , the coefficients of  $\mathbf{Y}(t)$  is defined as in the proposition will be smooth under the lifted frame, meaning that  $\mathbf{Y}(t)$  is smooth around  $t$ . Since  $t$  is arbitrary, we have proved that  $\mathbf{Y}(t)$  is a smooth vector field along  $c$ .

Given that the notations and language are slightly different in [Michor, 2008](#), we provide a bit more explanation for reader's convenience. We apply the two theorems for the fiber bundle  $(E, p, M, S)$  with  $E = \Theta$ ,  $p = \pi$ ,  $S = O(q)$ , and  $M$  is just our  $M$ . The map  $\Phi$  mentioned in their text (called the Ehresmann connection), merely assigns each  $\theta \in \Theta$  the linear projection map  $\Phi_\theta : T_\theta\Theta \rightarrow T_\theta\Theta$  onto the subspace  $V_\theta$ . Given the smooth curve  $\gamma$  in the statement of our proposition, Theorem 17.8 implemented with  $x = \gamma(t_0)$  and  $u_x = (\mathbf{A}, s)$  produces a map  $\text{Pt}_\gamma$ . We set  $c(t) = \text{Pt}_\gamma((\mathbf{A}, s), t)$ , which is a smooth curve. Property (1) of Theorem 17.8 shows that  $c(t_0) = (\mathbf{A}, s)$  and  $\pi(c(t)) = \gamma(t)$  for all  $t$  such that  $c$  is well-defined. Property (2) of Theorem 17.8 shows that  $c'(t) \in H_{c(t)}$  for all  $t$  such that  $c$  is well-defined. Given that  $S = O(q)$  is compact, Theorem 17.9 shows that  $c$  is defined on the entire  $(a, b)$ . An application of chain rule from  $\pi \circ c = \gamma$  shows that  $d\pi_{c(t)}(c'(t)) = \gamma'(t)$  for all  $t \in (a, b)$ . Finally, as pointed out in the first proof of Theorem 17.8,  $\text{Pt}_\gamma$  is unique, or equivalently  $c$  is unique.  $\square$

## 1.2.2 Geodesics, the Exponential Map, and the Logarithmic Map

Here is a classification result about all geodesics in  $M$ .

**Proposition 5.** Given  $[\mathbf{A}, s] \in M$  and  $\mathbf{v} \in T_{[\mathbf{A}, s]}M$ , all geodesics in  $M$  starting at  $[\mathbf{A}, s]$  with initial velocity  $\mathbf{v}$  takes the format of curves  $\gamma : I \rightarrow M$ , where  $I$  is an interval containing 0 in its interior, defined by

$$\gamma(t) = [\mathbf{A} + t\mathbf{V}, s + tc], \quad (1.14)$$

where  $(\mathbf{V}, c) \in H_{(\mathbf{A}, s)}$  is the horizontal lift of  $\mathbf{v}$  at  $(\mathbf{A}, s)$ .

*Proof.* First of all, the curve  $\gamma$  given by (1.14) is well defined. If  $(\mathbf{A}\mathbf{Q}, s)$ , where  $\mathbf{Q} \in O(q)$ , is another representative of  $[\mathbf{A}, s]$ , let  $(\mathbf{V}', c')$  be the horizontal lift of  $\mathbf{v}$  to  $(\mathbf{A}\mathbf{Q}, s)$ . With the function  $R_{\mathbf{Q}} : \Theta \rightarrow \Theta$  defined by  $R_{\mathbf{Q}}((\mathbf{A}, s)) = (\mathbf{A}\mathbf{Q}, s)$ , we have

$$\mathbf{v} = d\pi_{(\mathbf{A}, s)}(\mathbf{V}, c) = d(\pi \circ R_{\mathbf{Q}})_{(\mathbf{A}, s)}(\mathbf{V}, c) = d\pi_{(\mathbf{A}\mathbf{Q}, s)}(dR_{\mathbf{Q}})_{(\mathbf{A}, s)}(\mathbf{V}, c) = d\pi_{(\mathbf{A}\mathbf{Q}, s)}(\mathbf{V}\mathbf{Q}, c).$$

Thus,  $\mathbf{V}' = \mathbf{V}\mathbf{Q}$  and  $c' = c$ . Substitute this into the definition of  $\gamma$  will yield the exact same curve.

Given that  $\gamma$  is well-defined, it must be that  $(\mathbf{A} + t\mathbf{V}, s + tc) \in \Theta$  for all  $t \in I$ . The curve  $c : I \rightarrow \Theta$  with

$$c(t) = (\mathbf{A} + t\mathbf{V}, s + tc) \quad (1.15)$$

for all  $t \in I$  is then well-defined. Being a straight line in Euclidean space,  $c$  is a geodesic. Moreover, it has horizontal initial velocity and satisfies  $\pi \circ c = \gamma$ . By Corollary 26.12 from [Michor, 2008](#),  $\gamma$  must be a geodesic. Conversely, suppose that  $\gamma : I \rightarrow M$  is a geodesic in  $M$  starting at  $[\mathbf{A}, s]$  with initial velocity  $\mathbf{v}$ . Let  $c$  be the horizontal lift of  $\gamma$  at 0 and representative  $(\mathbf{A}, s)$  given by Proposition 4. Then  $c$  starts at  $(\mathbf{A}, s)$  and has its initial

velocity  $(\mathbf{V}, c)$  being the horizontal lift of  $\mathbf{v}$  at  $(\mathbf{A}, s)$ . Lemma 26.11 in [Michor, 2008](#) shows that  $c$  is a geodesic in  $\Theta$ . Given that straight lines are the only geodesics in Euclidean spaces,  $c$  must be defined by (1.15) for all  $t \in I$  and  $\gamma = \pi \circ c$  must take the format (1.14).  $\square$

We now move to compute the exponential map and the logarithmic map. The following two lemmas are crucial for computing the orthogonal matrix based on given points in  $M$ .

**Lemma 6.** If  $\mathbf{L}$  is an invertible  $q \times q$  matrix and  $\mathbf{Q} \in O(q)$  are such that  $\mathbf{LQ}$  is symmetric, then there exists matrices  $\mathbf{R} \in O(q)$ ,  $\mathbf{P} \in O(q)$ , a  $q \times q$  diagonal matrix  $\mathbf{D}$  whose diagonal entries are 1 or  $-1$ , and a  $q \times q$  diagonal matrix  $\mathbf{S}$  with diagonal entries  $\sigma_1(\mathbf{L}) \geq \dots \geq \sigma_q(\mathbf{L}) > 0$  such that  $\mathbf{L} = \mathbf{RSP}^T$  and  $\mathbf{Q} = \mathbf{PDR}^T$ .

If we add furthermore the assumption that  $\mathbf{L}$  is symmetric positive definite, then we can achieve the results mentioned above with  $\mathbf{P} = \mathbf{R}$ . In particular, this implies  $\mathbf{Q} = \mathbf{Q}^T$  and  $\|\mathbf{Q} - \mathbf{I}_q\| = \|\mathbf{D} - \mathbf{I}_q\|$  can take only 0 or 2 as values.

*Proof.* Let us start by applying the singular value decomposition of  $\mathbf{L}$  to get matrices  $\mathbf{R}' \in O(q)$ ,  $\mathbf{P}' \in O(q)$  and a diagonal matrix  $\mathbf{S}$  with diagonal entries  $\sigma_1(\mathbf{L}) \geq \dots \geq \sigma_q(\mathbf{L}) > 0$  such that  $\mathbf{L} = \mathbf{R}'\mathbf{S}(\mathbf{P}')^T$ . Note that all singular values are positive because  $\mathbf{L}$  is invertible. Substituting the given condition that  $\mathbf{LQ}$  is symmetric will yield that  $\mathbf{SJ} = \mathbf{J}^T\mathbf{S}$ , where  $\mathbf{J} := (\mathbf{P}')^T\mathbf{Q}\mathbf{R}'$ .

Clearly  $\mathbf{J} \in O(q)$ . The equation

$$\mathbf{S}^2 = \mathbf{SJJ}^T\mathbf{S} = (\mathbf{SJ})(\mathbf{SJ})^T = (\mathbf{J}^T\mathbf{S})(\mathbf{J}^T\mathbf{S})^T = \mathbf{J}^T\mathbf{S}^2\mathbf{J}$$

shows that

$$\mathbf{JS}^2 = \mathbf{S}^2\mathbf{J}.$$

Looking at each entry of both side, we end up with

$$\mathbf{J}_{ij}(\sigma_j(\mathbf{L})^2 - \sigma_i(\mathbf{L})^2) = 0$$

for  $i, j = 1, \dots, q$ . Given that all singular values are positive, we in fact have

$$\mathbf{J}_{ij}(\sigma_j(\mathbf{L}) - \sigma_i(\mathbf{L})) = 0 \tag{1.16}$$

for all  $i$  and all  $j$  so that  $\mathbf{JS} = \mathbf{SJ} = \mathbf{J}^T\mathbf{S}$ . Inverting  $\mathbf{S}$  yields that  $\mathbf{J}$  is actually symmetric. Being a symmetric orthogonal matrix,  $\mathbf{J}^2 = \mathbf{I}_q$  and the eigenvalues of  $\mathbf{J}$  must be 1 or  $-1$ . Another look at (1.16) shows that  $\mathbf{J}$  is a block-diagonal matrix, diagonal if all singular values are distinct. Furthermore, each block of  $\mathbf{J}$  must still be orthogonal and symmetric.

We now diagonalize  $\mathbf{J}$  based on the above information. Suppose that  $\mathbf{S} = \bigoplus_{i=1}^k s_i \mathbf{I}_{m_i}$ , where  $s_1 > \dots > s_k$  are the unique diagonal entries of  $\mathbf{S}$  and each repeats for  $m_i$  times. Thus,  $\mathbf{J}$  has  $k$  blocks. Write the  $i$ th block by  $\mathbf{J}_i$  so that  $\mathbf{J} = \bigoplus_{i=1}^k \mathbf{J}_i$ . The spectral theorem then allows us to orthogonally diagonalize  $\mathbf{J}_i = \mathbf{K}_i \mathbf{D}_i \mathbf{K}_i^T$ , where  $\mathbf{K}_i \in O(m_i)$  and  $\mathbf{D}_i$  is an  $m_i \times m_i$  diagonal matrix with all diagonal entries being 1 or  $-1$ . Put  $\mathbf{K} = \bigoplus_{i=1}^k \mathbf{K}_i$  and  $\mathbf{D} = \bigoplus_{i=1}^k \mathbf{D}_i$ . Then  $\mathbf{K} \in O(q)$ ,  $\mathbf{D}$  is a  $q \times q$  diagonal matrix with 1 or  $-1$  diagonal entry, and  $\mathbf{J} = \mathbf{KDK}^T$ . This involved construction leads us to the outcome that  $\mathbf{K}$  in fact commutes with  $\mathbf{S}$ :

$$\mathbf{KS} = \bigoplus_{i=1}^k s_i \mathbf{K}_i = \mathbf{SK}.$$

Put  $\mathbf{P} = \mathbf{P}'\mathbf{K}$ ,  $\mathbf{R} = \mathbf{R}'\mathbf{K}$ . We have

$$\mathbf{Q} = \mathbf{P}'\mathbf{J}(\mathbf{R}')^T = \mathbf{P}'\mathbf{KDK}^T(\mathbf{R}')^T = \mathbf{PDR}^T$$

and

$$\mathbf{RSP}^T = \mathbf{R}'\mathbf{KSK}^T(\mathbf{P}')^T = \mathbf{R}'\mathbf{SKK}^T(\mathbf{P}')^T = \mathbf{R}'\mathbf{S}(\mathbf{P}')^T = \mathbf{L}.$$

This completes the proof.

Finally, let us just note that, in case  $\mathbf{L}$  is symmetric positive definite, the spectral theorem allows us to take  $\mathbf{R}' = \mathbf{P}'$  from the start and it will yield in the end  $\mathbf{L} = \mathbf{PSP}^T$  and  $\mathbf{Q} = \mathbf{PDP}^T$ . It then follows easily that  $\mathbf{Q}^2 = \mathbf{I}_q$  and  $\mathbf{Q}^T = \mathbf{Q}$ . If  $\mathbf{D} = \mathbf{I}_q$ , then  $\|\mathbf{Q} - \mathbf{I}_q\| = \|\mathbf{D} - \mathbf{I}_q\| = 0$ . If any entry of  $\mathbf{D}$  is  $-1$ ,  $\|\mathbf{D} - \mathbf{I}_q\| = 2$ .  $\square$

**Lemma 7.** If  $\mathbf{L}$  is an invertible  $q \times q$  matrix with singular value decomposition  $\mathbf{L} = \mathbf{RSP}^T$ , then

$$\mathbf{PR}^T = (\mathbf{L}^T \mathbf{L})^{-\frac{1}{2}} \mathbf{L}^T.$$

*Proof.* Given that

$$\mathbf{L}^T \mathbf{L} = \mathbf{PS}^2 \mathbf{P}^T,$$

we have

$$(\mathbf{L}^T \mathbf{L})^{\frac{1}{2}} = \mathbf{PSP}^T.$$

Therefore:

$$\mathbf{L} = \mathbf{RP}^T \mathbf{PSP}^T = \mathbf{RP}^T (\mathbf{L}^T \mathbf{L})^{\frac{1}{2}}.$$

The result follows from inverting  $(\mathbf{L}^T \mathbf{L})^{\frac{1}{2}}$  and take transpose.  $\square$

We now state and prove all the results about the exponential and the logarithmic map in one single proposition.

**Proposition 8.** The exponential map is computed as

$$\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}) = [\mathbf{A} + \mathbf{V}, s + c], \quad (1.17)$$

where  $[\mathbf{A}, s] \in M$ ,  $\mathbf{v} \in T_{[\mathbf{A}, s]}M$ , and  $(\mathbf{V}, c)$  is the horizontal lift of  $\mathbf{v}$  at  $(\mathbf{A}, s)$ .

Let us use  $\mathcal{E}_{[\mathbf{A}, s]}$  to denote the open ball in  $T_{[\mathbf{A}, s]}M \cong \mathbb{R}^d$  centred at  $\mathbf{0}$  of radius  $\min(\sigma_{\min}(\mathbf{A}), s)$ . Given that (1.17) is well-defined at least over all  $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$ , we from now on treat  $\mathcal{E}_{[\mathbf{A}, s]}$  as the domain of the exponential map. That is, the function  $\text{Exp}_{[\mathbf{A}, s]} : \mathcal{E}_{[\mathbf{A}, s]} \rightarrow U_{[\mathbf{A}, s]}$ , where  $U_{[\mathbf{A}, s]} := \text{Exp}_{[\mathbf{A}, s]}(\mathcal{E}_{[\mathbf{A}, s]})$  is an open subset of  $M$  referred to as a geodesic ball, is defined by (1.17). Note the following facts:

1. For every  $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$ ,  $\mathbf{A}^T(\mathbf{A} + \mathbf{V})$  is symmetric positive definite.
2.  $\text{Exp}_{[\mathbf{A}, s]}$  is a diffeomorphism.
3. If  $[\mathbf{B}, t] \in U_{[\mathbf{A}, s]}$ , then for every representative  $(\mathbf{A}, s)$  of  $[\mathbf{A}, s]$  and every representative  $(\mathbf{B}, t)$  of  $[\mathbf{B}, t]$ , the matrix  $\mathbf{A}^T \mathbf{B}$  will always be invertible. Given representative  $(\mathbf{A}, s)$  of  $[\mathbf{A}, s]$ , some representative  $(\mathbf{B}, t)$  of  $[\mathbf{B}, t]$  will furthermore make  $\mathbf{A}^T \mathbf{B}$  symmetric positive definite.
4. The logarithmic map  $\text{Exp}_{[\mathbf{A}, s]}^{-1} : U_{[\mathbf{A}, s]} \rightarrow \mathcal{E}_{[\mathbf{A}, s]}$  is computed as follows: for every  $[\mathbf{B}, t] \in U_{[\mathbf{A}, s]}$ ,

$$\text{Exp}_{[\mathbf{A}, s]}^{-1}([\mathbf{B}, t]) = d\pi_{(\mathbf{A}, s)}((\mathbf{BQ}(\mathbf{A}, \mathbf{B}) - \mathbf{A}, t - s)), \quad (1.18)$$

where

$$\mathbf{Q}(\mathbf{A}, \mathbf{B}) = (\mathbf{B}^T \mathbf{A} \mathbf{A}^T \mathbf{B})^{-\frac{1}{2}} \mathbf{B}^T \mathbf{A} \in O(q). \quad (1.19)$$

5. The geodesic ball  $U_{[\mathbf{A}, s]}$  equals to the set of all  $[\mathbf{B}, t] \in M$  such that  $\mathbf{A}^T \mathbf{B}$  is invertible and

$$\|\mathbf{BQ}(\mathbf{A}, \mathbf{B}) - \mathbf{A}\|_F^2 + (t - s)^2 < \min(\sigma_{\min}(\mathbf{A}), s)^2. \quad (1.20)$$

6. If  $[\mathbf{B}, t] \in U_{[\mathbf{A}, s]}$ , then the Riemannian distance between  $[\mathbf{A}, s]$  and  $[\mathbf{B}, t]$  equals to

$$\left| \text{Exp}_{[\mathbf{A}, s]}^{-1}([\mathbf{B}, t]) \right| = \sqrt{\|\mathbf{B}\mathbf{Q}(\mathbf{A}, \mathbf{B}) - \mathbf{A}\|_F^2 + (t - s)^2}, \quad (1.21)$$

which is realized by one unique geodesic  $\gamma : [0, 1] \rightarrow M$  defined by, for all  $u \in [0, 1]$ :

$$\gamma(u) = \text{Exp}_{[\mathbf{A}, s]}(u \text{Exp}_{[\mathbf{A}, s]}^{-1}([\mathbf{B}, t])). \quad (1.22)$$

*Proof.* Before we start the proof, let us first check that all the statements are indeed well-defined. If  $\mathbf{R} \in O(q)$  and  $\mathbf{P} \in O(q)$ , then

$$\mathbf{Q}(\mathbf{A}\mathbf{P}, \mathbf{B}\mathbf{R}) = \mathbf{R}^T \mathbf{Q}(\mathbf{A}, \mathbf{B}) \mathbf{P}. \quad (1.23)$$

Therefore, the same computation as in the proof of Proposition 5 will show that (1.18) is indeed well-defined. Also, the contents of the inequality (1.20) and the expression (1.21) are independent of representatives and the description of the set  $U_{[\mathbf{A}, s]}$  makes sense.

Proposition 5 already shows that the computation of the exponential map is valid as long as  $(\mathbf{A} + \mathbf{V}, s + c) \in \Theta$ . Therefore, the exponential map is certainly well-defined over all  $\mathbf{v}$  with small enough norm. To have a quantified upper bound of the norm, we use the classical result of Weyl's perturbation inequality (see for example Chapter 7.3 Problem 16 equation (7.3.15) of Horn and Johnson, 2013) to get

$$\sigma_{\min}(\mathbf{A} + \mathbf{V}) \geq \sigma_{\min}(\mathbf{A}) - \|\mathbf{V}\| \geq \sigma_{\min}(\mathbf{A}) - \|\mathbf{V}\|_F \geq \sigma_{\min}(\mathbf{A}) - |\mathbf{v}| > 0,$$

whenever  $|\mathbf{v}| < \min(\sigma_{\min}(\mathbf{A}), s)$  or equivalently  $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$ . We have made use of the two facts that  $\|\mathbf{V}\| \leq \|\mathbf{V}\|_F$  for any matrix  $\mathbf{V}$  and that  $|\mathbf{v}| = \sqrt{\|\mathbf{V}\|_F^2 + c^2}$  by isometry. Thus, for all  $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$ ,  $\text{rank}(\mathbf{A} + \mathbf{V}) = q$  and the exponential map is well-defined in  $\mathcal{E}_{[\mathbf{A}, s]}$ .

We now start the proof of the five facts of the function  $\text{Exp}_{[\mathbf{A}, s]} : \mathcal{E}_{[\mathbf{A}, s]} \rightarrow U_{[\mathbf{A}, s]}$ .

The first fact can be quickly settled. Let  $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$  with horizontal lift  $(\mathbf{V}, c)$  to  $(\mathbf{A}, s)$  be given. As proved in Proposition 1,  $\mathbf{V}^T \mathbf{A} = \mathbf{A}^T \mathbf{V}$  and thus  $\mathbf{A}^T(\mathbf{A} + \mathbf{V})$  is symmetric. If  $\mathbf{x} \in \mathbb{R}^q$  is any unit vector, then

$$\mathbf{x}^T \mathbf{A}^T(\mathbf{A} + \mathbf{V})\mathbf{x} = |\mathbf{A}\mathbf{x}|^2 + (\mathbf{A}\mathbf{x})^T \mathbf{V}\mathbf{x} \geq |\mathbf{A}\mathbf{x}|(|\mathbf{A}\mathbf{x}| - |\mathbf{V}\mathbf{x}|) \geq \sigma_{\min}(\mathbf{A})(\sigma_{\min}(\mathbf{A}) - \|\mathbf{V}\|_F) > 0$$

because  $\sigma_{\min}(\mathbf{A}) > |\mathbf{v}| \geq \|\mathbf{V}\|_F$ .

The second fact has a longer proof. Being a well-defined exponential map,  $\text{Exp}_{[\mathbf{A}, s]}$  is automatically a smooth map. To prove that it is actually a diffeomorphism in  $\mathcal{E}_{[\mathbf{A}, s]}$ , we must first show that it is injective in  $\mathcal{E}_{[\mathbf{A}, s]}$  so that the inverse function, the logarithmic map  $\text{Exp}_{[\mathbf{A}, s]}^{-1}$ , exists. Once this is done, we prove that  $U_{[\mathbf{A}, s]}$  is open and that  $\text{Exp}_{[\mathbf{A}, s]}^{-1}$  is smooth in one shot by proving that its differential  $d\left(\text{Exp}_{[\mathbf{A}, s]}\right)_v$  is an isomorphism for every given  $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$ .

We thus move to show that  $\text{Exp}_{[\mathbf{A}, s]}$  is injective in  $\mathcal{E}_{[\mathbf{A}, s]}$ . Suppose that  $\mathbf{v}_1$  and  $\mathbf{v}_2$  are two vectors in  $\mathcal{E}_{[\mathbf{A}, s]}$  who lifts to  $(\mathbf{V}_1, c_1)$  and  $(\mathbf{V}_2, c_2)$  respectively at  $(\mathbf{A}, s)$  such that  $\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}_1) = \text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}_2)$ . We prove that  $\mathbf{v}_1 = \mathbf{v}_2$ . (1.17) immediately shows that  $c_1 = c_2$  and

$$\mathbf{A} + \mathbf{V}_1 = (\mathbf{A} + \mathbf{V}_2)\mathbf{Q}$$

for some  $\mathbf{Q} \in O(q)$ . Lemma 6 with the symmetric positive definite matrix  $\mathbf{L} = \mathbf{A}^T(\mathbf{A} + \mathbf{V}_2)$  shows that  $\|\mathbf{Q} - \mathbf{I}_q\|$  can only equal to 0 or 2. Given that

$$\sigma_{\min}(\mathbf{A})\|\mathbf{Q} - \mathbf{I}_q\| \leq \|\mathbf{A}\mathbf{Q} - \mathbf{A}\| = \|\mathbf{V}_1 - \mathbf{V}_2\mathbf{Q}\| \leq \|\mathbf{V}_1\| + \|\mathbf{V}_2\| < 2\sigma_{\min}(\mathbf{A}),$$

we have proved that  $\|\mathbf{Q} - \mathbf{I}_q\| < 2$  and  $\mathbf{Q} = \mathbf{I}_q$ . Therefore,  $\mathbf{V}_1 = \mathbf{V}_2$  and  $\mathbf{v}_1 = \mathbf{v}_2$ . The exponential map is indeed injective in  $\mathcal{E}_{[\mathbf{A}, s]}$ .

Next, let  $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$  be given. We prove that  $d(\text{Exp}_{[\mathbf{A}, s]})_{\mathbf{v}}$  is an isomorphism because  $\ker(d(\text{Exp}_{[\mathbf{A}, s]})_{\mathbf{v}})$  consists only of the zero vector. Suppose that  $\mathbf{v}$  lifts to  $(\mathbf{V}, c)$  at  $(\mathbf{A}, s)$ . To make the task notationally feasible, we implicitly canonically identify the Euclidean spaces together. Let us write  $l : T_{[\mathbf{A}, s]}M \rightarrow H_{(\mathbf{A}, s)}$  for the horizontal lift map:  $l(\mathbf{v}) = (\mathbf{V}, c)$ , which is a linear isometry. Define a map  $F : l(\mathcal{E}_{[\mathbf{A}, s]}) \rightarrow M$  by setting, for any  $(\mathbf{U}, b) \in l(\mathcal{E}_{[\mathbf{A}, s]}) \subseteq H_{(\mathbf{A}, s)}$ , that  $F((\mathbf{U}, b)) = \pi((\mathbf{A} + \mathbf{U}, s + b))$ . In this notation,  $\text{Exp}_{[\mathbf{A}, s]} = F \circ l$  and the chain rule shows that

$$d(\text{Exp}_{[\mathbf{A}, s]})_{\mathbf{v}} = dF_{(\mathbf{V}, c)} \circ dl_{\mathbf{v}}.$$

Therefore, the only thing left to show is that  $\ker(dF_{(\mathbf{V}, c)})$  contains only the zero vector. For any  $(\mathbf{U}, b) \in l(\mathcal{E}_{[\mathbf{A}, s]})$ , we always have

$$dF_{(\mathbf{V}, c)}((\mathbf{U}, b)) = \left. \frac{d}{dt} \right|_{t=0} \pi((\mathbf{A} + \mathbf{V} + t\mathbf{U}, s + c + tb)) = d\pi_{(\mathbf{A} + \mathbf{V}, s + c)}((\mathbf{U}, b)).$$

Therefore, if  $(\mathbf{U}, b) \in \ker(dF_{(\mathbf{V}, c)})$ , then  $(\mathbf{U}, b) \in V_{(\mathbf{A} + \mathbf{V}, s + c)}$  so that  $b = 0$  and  $\mathbf{U} = (\mathbf{A} + \mathbf{V})\Omega$  for some  $q \times q$  skew-symmetric matrix  $\Omega$ . But  $(\mathbf{U}, b) \in H_{(\mathbf{A}, s)}$  as well. Thus,  $\mathbf{A}^T(\mathbf{A} + \mathbf{V})\Omega = \Omega^T(\mathbf{A} + \mathbf{V})^T\mathbf{A}$ . Given that  $\mathbf{A}^T(\mathbf{A} + \mathbf{V})$  is also symmetric, we arrive at

$$\mathbf{A}^T(\mathbf{A} + \mathbf{V})\Omega + \Omega\mathbf{A}^T(\mathbf{A} + \mathbf{V}) = \mathbf{0}.$$

Lemma 2 with  $\mathbf{S} = \mathbf{A}^T(\mathbf{A} + \mathbf{V})$  forces  $\Omega = \mathbf{0}$ . We have proven that  $\text{Exp}_{[\mathbf{A}, s]}$  is a diffeomorphism in  $\mathcal{E}_{[\mathbf{A}, s]}$  and that  $U_{[\mathbf{A}, s]}$  is an open subset of  $M$ .

Now that the logarithmic map  $\text{Exp}_{[\mathbf{A}, s]}^{-1} : U_{[\mathbf{A}, s]} \rightarrow \mathcal{E}_{[\mathbf{A}, s]}$  is well-defined and is smooth, we will compute it and show the next three facts in one shot. Let  $[\mathbf{B}, t] \in U_{[\mathbf{A}, s]}$  be given. By (1.17),  $[\mathbf{B}, t] = [\mathbf{A} + \mathbf{V}, s + c]$ , where  $(\mathbf{V}, c)$  is the horizontal lift of  $\mathbf{v} = \text{Exp}_{[\mathbf{A}, s]}^{-1}([\mathbf{B}, t])$  to  $(\mathbf{A}, s)$ . It immediately follows that  $c = t - s$ . The first fact shows that, with  $\mathbf{B}_0 := \mathbf{A} + \mathbf{V}$ ,  $\mathbf{A}^T\mathbf{B}_0$  is symmetric positive definite. Given that the value of  $|\det(\mathbf{A}^T\mathbf{B})|$  is independent of representatives,  $\mathbf{A}^T\mathbf{B}$  is always invertible for any representative. The third fact is therefore established. Given that  $\mathbf{Q}(\mathbf{A}, \mathbf{B}_0) = \mathbf{I}_q$ , we immediately see that (1.18) holds with  $\mathbf{B}_0$  in place of  $\mathbf{B}$ . However, (1.18) is independent of representatives so it is valid for every representative  $\mathbf{B}$ . Finally, let us write  $C$  for the set of all points  $[\mathbf{B}, t] \in M$  such that  $\mathbf{A}^T\mathbf{B}$  is invertible and satisfies (1.20). By using  $\mathbf{B}_0$  in place of  $\mathbf{B}$  in (1.20), it is clear that  $U_{[\mathbf{A}, s]} \subseteq C$ . Conversely, let  $[\mathbf{B}, t] \in C$  be given. Put  $\mathbf{B}_0 := \mathbf{B}\mathbf{Q}(\mathbf{A}, \mathbf{B})$ . Then  $\mathbf{A}^T\mathbf{B}_0$  is again symmetric positive definite. In particular,  $\mathbf{A}^T(\mathbf{B}_0 - \mathbf{A})$  is symmetric and  $(\mathbf{V}, c) := (\mathbf{B}_0 - \mathbf{A}, t - s)$  is a horizontal vector at  $(\mathbf{A}, s)$ . If  $\mathbf{v} := d\pi_{(\mathbf{A}, s)}((\mathbf{V}, c))$ , then (1.20) shows that  $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$  and  $[\mathbf{B}, t] \in U_{[\mathbf{A}, s]}$ . We conclude that  $U_{[\mathbf{A}, s]} = C$ .

Finally, let us calculate the Riemannian distance between  $[\mathbf{B}, t] \in U_{[\mathbf{A}, s]}$  and  $[\mathbf{A}, s]$ . Proposition 6.11 in Lee (2018) shows that the Riemannian distance between points in a geodesic ball is indeed attained by exactly one geodesic. It suffices to find a geodesic minimizing the Riemannian distance among all geodesics. By Proposition 5, all geodesics connecting  $[\mathbf{A}, s]$  to  $[\mathbf{B}, t]$  takes the form  $\gamma(u) = [\mathbf{A} + u\mathbf{V}, s + uc]$ , where  $(\mathbf{V}, c)$  is the horizontal lift of some vector  $\mathbf{v} \in T_{[\mathbf{A}, s]}M$ . It is clear that  $c = t - s$  and  $\mathbf{V} = \mathbf{B}\mathbf{Q} - \mathbf{A}$  for some  $\mathbf{Q} \in O(q)$ . Given that the endpoints are fixed, the time is fixed, and  $\gamma$  has constant speed  $|\mathbf{v}|$ , the choice of  $\mathbf{v}$  with minimal speed will give the smallest arclength and hence the smallest Riemannian distance. To carry out this minimization, Lemma 6 shows that there exists  $\mathbf{R} \in O(q)$ ,  $\mathbf{P} \in O(q)$  such that  $\mathbf{A}^T\mathbf{B} = \mathbf{R}\mathbf{S}\mathbf{P}^T$  and  $\mathbf{Q} = \mathbf{P}\mathbf{D}\mathbf{R}^T$ . Let us compute the norm of  $\mathbf{v}$ :

$$\begin{aligned} g_{[\mathbf{A}, s]}(\mathbf{v}, \mathbf{v}) &= g_{(\mathbf{A}, s)}^{\Theta}((\mathbf{B}\mathbf{Q} - \mathbf{A}, t - s), (\mathbf{B}\mathbf{Q} - \mathbf{A}, t - s)) \\ &= \text{tr}((\mathbf{B}\mathbf{Q} - \mathbf{A})(\mathbf{B}\mathbf{Q} - \mathbf{A})^T) + (t - s)^2 = \text{tr}(\mathbf{B}\mathbf{B}^T) + \text{tr}(\mathbf{A}\mathbf{A}^T) - 2\text{tr}(\mathbf{S}\mathbf{D}) + (t - s)^2, \end{aligned}$$

where the last equality substituted  $\mathbf{A}^T\mathbf{B}\mathbf{Q}' = \mathbf{R}\mathbf{S}\mathbf{P}^T\mathbf{P}\mathbf{D}\mathbf{R}^T = \mathbf{R}\mathbf{S}\mathbf{D}\mathbf{R}^T$ . The fact that the diagonal entries of  $\mathbf{S}$  are strictly positive show that  $g_{[\mathbf{A}, s]}(\mathbf{v}, \mathbf{v})$  is minimized precisely when  $\mathbf{D} = \mathbf{I}_q$ . By Lemma 7, it must be that  $\mathbf{Q} = \mathbf{P}\mathbf{R}^T = \mathbf{Q}(\mathbf{A}, \mathbf{B})$  and the last fact holds. The proof is complete.  $\square$

**Remark 9.** In the above proof, it might seem that the proof of (1.18) is trivialized while the amazing matrix (1.19) seems to come from nowhere. This is not at all the case. Originally, the matrix  $Q(A, B)$  comes from considering the cases for a generic representative whose matrix  $A^T B$  is only invertible, but not necessarily symmetric or positive definite. Given  $[B, t] \in U_{[A, s]}$ , (1.17) shows that

$$\text{Exp}_{[A, s]}^{-1}([B, t]) = d\pi_{(A, s)}((BQ - A, t - s))$$

for some  $Q \in O(q)$ . Given that  $A^T(BQ - A)$  is symmetric, it follows that  $A^T BQ$  is symmetric and Lemma 6 shows that there exists  $R \in O(q)$ ,  $P \in O(q)$  such that  $A^T B = RSP^T$  and  $Q = PDR^T$ . Thus, by continuity of the logarithmic map, if  $[B, t] \rightarrow [A, s]$  and for some representative  $(B, t) \rightarrow (A, s)$ , then it must be the case that  $BQ - A \rightarrow 0$  or equivalently  $Q \rightarrow I_q$ . Therefore, heuristically, the only reasonable choice of  $D$  is  $I_q$ . So  $Q = PR^T$ . This value is actually uniquely determined by  $A$  and  $B$  and gives exactly  $Q = Q(A, B)$ , as shown in Lemma 7, despite that neither  $P$  nor  $R$  is unique. This argument is not discarded. In the proof of injectivity of the exponential map, the argument is present but simplified by symmetric positive definiteness. The problem of calculating the Riemannian distance actually makes us show a stronger result that minimizing the norm of  $BQ - A$  with fixed  $A$  and  $B$  will already lead to  $Q = Q(A, B)$ .

### 1.2.3 Parallel Transport

In Sun et al. (2026), the parallel transport maps are always used along the curve minimizing Riemannian distance. (In their work, the underlying manifold is always assumed to be complete and therefore such a parallel transport map is always well-defined. Given that our manifolds  $\Theta$  and  $M$  are both not complete, it is not immediately clear that a parallel transport map along distance minimizing curve should be well-defined.) We will make the same definitions but only for given  $[A, s] \in M$  and  $[B, t] \in U_{[A, s]}$  because it is only in this case where we have shown that there exists a unique distance minimizing curve  $\gamma$ , given in (1.22). We will also compute the parallel transport vector field and hence the parallel transport map. (Although, in the current low dimensional document, the only thing we will use out of this computation is that the parallel transport map depends *smoothly on the tuple of all its entries*.) (I tried to trivialize this by searching for an existing result that proves the smoothness on all entries in textbooks, but no textbooks seem to have prove this exact statement and I have to do it in the harder way, namely to compute everything out and see the smoothness directly.) This is summarised in the following proposition.

**Proposition 10.** Given  $[A, s] \in M$  and  $[B, t] \in U_{[A, s]}$ , put  $v := \text{Exp}_{[A, s]}^{-1}([B, t])$ . Then, as already proved in Proposition 8, with  $V = BQ(A, B) - A$  and  $c = t - s$ ,  $(V, c)$  is the horizontal lift of  $v$  at  $(A, s)$  and  $\gamma : [0, 1] \rightarrow M$  defined by, for all  $u \in [0, 1]$ ,

$$\gamma(u) = [A + uV, s + uc],$$

is the unique distance minimizing curve connecting  $[A, s]$  to  $[B, t]$ , in accordance with (1.22). The lift of  $\gamma$  to  $(A, s)$  is the curve  $L : [0, 1] \rightarrow \Theta$  defined by, for all  $u \in [0, 1]$ ,

$$L(u) = (A + uV, s + uc).$$

Let us write  $\Pi_{[B, t]}^{[A, s]} : T_{[B, t]}M \rightarrow T_{[A, s]}M$  for the parallel transport map along  $\gamma$ . Given  $h \in T_{[B, t]}M$ , let  $(Z, \beta)$  be the lift of  $h$  at  $(BQ(A, B), t)$  and  $X$  be the parallel transport vector field along  $\gamma$  for  $h$ . This means  $X(u) \in T_{\gamma(u)}M$  for  $u \in [0, 1]$ ,  $X(1) = h$ ,  $X$  is smooth, and the covariant derivative along  $\gamma$ , denoted by  $\frac{D}{du}X(u)$ , always equal to the zero vector field along the curve. Then, for each  $u \in [0, 1]$ ,

$$X(u) = d\pi_{L(u)}((Z(u), \beta)) \tag{1.24}$$

and

$$\Pi_{[B, t]}^{[A, s]}(h) = X(0),$$

where  $Z(u)$ , a  $p \times q$  matrix, is the unique solution of the first order ordinary differential equation

$$\frac{d}{du} \mathbf{Z}(u) = - \frac{\partial}{\partial w} \Big|_{w=u} (\mathbf{A} + w\mathbf{V}) \psi_{(\mathbf{A}+w\mathbf{V})^T(\mathbf{A}+w\mathbf{V})}^{-1} ((\mathbf{A} + w\mathbf{V})^T \mathbf{Z}(u) - \mathbf{Z}(u)^T (\mathbf{A} + w\mathbf{V})) \quad (1.25)$$

subject to the initial condition  $\mathbf{Z}(1) = \mathbf{Z}$ . Here  $\psi$  is as in Lemma 2.

Finally, as a part of the conclusion, the quantity  $\Pi_{[\mathbf{B}, t]}^{[\mathbf{A}, s]}(\mathbf{h})$  depends smoothly on the tuple  $([\mathbf{A}, s], [\mathbf{B}, t], \mathbf{h})$ .

*Proof.* Let  $\mathbf{Y}(u) = (\mathbf{Z}(u), \beta(u))$  be the horizontal lift of  $\mathbf{X}(u)$ , which is a smooth horizontal vector field along  $L$ . We prove that  $\mathbf{Z}(u)$  will satisfy (1.25) and that  $\beta(u) = \beta$  for all  $u$  using the projection map computed in Proposition 3.

For each point  $(\mathbf{A}, s) \in \Theta$ , let us write  $P_{(\mathbf{A}, s)} : T_{(\mathbf{A}, s)}\Theta \rightarrow T_{(\mathbf{A}, s)}\Theta$  for the orthogonal projection onto the horizontal tangent space  $H_{(\mathbf{A}, s)}\Theta$ . Given that  $\mathbf{Y}(u)$  is a horizontal vector field, it is invariant under this projection, namely, for all  $u \in [0, 1]$ :

$$\mathbf{Y}(u) = P_{L(u)}(\mathbf{Y}(u)). \quad (1.26)$$

By O'Neill's formula (see for instance Lee (2018) equation 5.27), the Levi-Civita connection in  $M$  evaluated at two vector fields in  $M$  lifts horizontally to the Levi-Civita connection in  $\Theta$  evaluated at the horizontally lifted version of the same two vector fields plus a verticle term. Consequently, the horizontal lift of  $\frac{D}{du} \mathbf{X}(u)$  equals to  $\frac{D}{du} \mathbf{Y}(u)$  plus a verticle term. Given that  $\mathbf{X}(u)$  is the parallel transport vector field,  $\frac{D}{du} \mathbf{X}(u)$  is the zero vector field and its horizontal lift must also be the zero vector field. Therefore,  $\frac{D}{du} \mathbf{Y}(u)$  is a verticle vector field and

$$P_{L(u)}\left(\frac{D}{du} \mathbf{Y}(u)\right) = \mathbf{0}. \quad (1.27)$$

With this in mind, covariant differentiating both sides of (1.26) to get

$$\frac{d}{du}(\mathbf{Z}(u), \beta(u)) = \frac{D}{du} \mathbf{Y}(u) = \frac{D}{du} P_{L(u)}(\mathbf{Y}(u)) = \frac{d}{du} P_{L(u)}(\mathbf{Y}(u)), \quad (1.28)$$

where we have used the fact that the covariant derivative in Euclidean space is just the ordinary derivative. We proceed by expanding out (1.27) and (1.28) using the chain rule, taking care of the dependencies of  $P_{L(u)}(\mathbf{Y}(u))$  on  $u$  for both entries. Put

$$F(w, \mathbf{W}) := \mathbf{W} - (\mathbf{A} + w\mathbf{V}) \psi_{(\mathbf{A}+w\mathbf{V})^T(\mathbf{A}+w\mathbf{V})}^{-1} ((\mathbf{A} + w\mathbf{V})^T \mathbf{W} - \mathbf{W}^T (\mathbf{A} + w\mathbf{V})).$$

Then  $F$  is smooth by Lemma 2, linear in the second argument, and, by (1.13),

$$P_{L(u)}(\mathbf{Y}(u)) = (F(u, \mathbf{Z}(u)), \beta(u)).$$

(1.27) now becomes

$$\mathbf{0} = P_{L(u)}\left(\frac{d}{du} \mathbf{Y}(u)\right) = \left(F\left(u, \frac{d}{du} \mathbf{Z}(u)\right), \frac{d}{du} \beta(u)\right). \quad (1.29)$$

Looking at the second component, we get that  $\frac{d}{du} \beta(u) = 0$  and that  $\beta(u)$  is constant. Next, the first component in (1.28) becomes

$$\begin{aligned} \frac{d}{du} \mathbf{Z}(u) &= \frac{d}{du} F(u, \mathbf{Z}(u)) = \frac{\partial F}{\partial w}(u, \mathbf{Z}(u)) + F\left(u, \frac{d}{du} \mathbf{Z}(u)\right) \\ &= \frac{\partial F}{\partial w}(u, \mathbf{Z}(u)) = - \frac{\partial}{\partial w} \Big|_{w=u} (\mathbf{A} + w\mathbf{V}) \psi_{(\mathbf{A}+w\mathbf{V})^T(\mathbf{A}+w\mathbf{V})}^{-1} ((\mathbf{A} + w\mathbf{V})^T \mathbf{Z}(u) - \mathbf{Z}(u)^T (\mathbf{A} + w\mathbf{V})). \end{aligned}$$

In the second term of the second equality, we have used the fact that  $F$  is linear in  $\mathbf{W}$  so that its derivative is the map itself. This second term equals to zero by the first component of (1.29).

We have therefore proved (1.24) and (1.25). Given that (1.25) is a system of first order differential equation with an initial condition with all dependencies being smooth, a standard result in the theory of ordinary differential equations show that  $\mathbf{Z}(u)$  exists, is unique, and depends smoothly on the tuple of everything in the equation and the initial condition. That is, for any  $u \in [0, 1]$ ,  $\mathbf{Z}(u)$  depends smoothly on the tuple  $([\mathbf{A}, s], [\mathbf{B}, t], \mathbf{Z}, \beta)$ . Given that horizontal lifts are smooth,  $\Pi_{[\mathbf{B}, t]}^{[\mathbf{A}, s]}(\mathbf{h})$  depends smoothly on the tuple  $([\mathbf{A}, s], [\mathbf{B}, t], \mathbf{h})$ , where different tangent spaces are identified together as the subspaces of  $\mathbb{R}^{p \times q+1}$ . This completes the proof.  $\square$

## 1.2.4 A Limit Result

We end this section with a result concerning the limit of a sequence.

**Proposition 11.** Given a sequence  $\{\mathbf{A}_n\}_{n=1}^\infty$  of real  $p \times q$  matrices with the property  $\mathbf{A}_n \mathbf{A}_n^T \rightarrow \mathbf{A} \mathbf{A}^T$  as  $n \rightarrow \infty$  for a  $p \times q$  matrix  $\mathbf{A}$  with  $\text{rank}(\mathbf{A}) = q$ , then

$$\lim_{n \rightarrow \infty} \min_{\mathbf{Q} \in O(q)} \|\mathbf{A}_n \mathbf{Q} - \mathbf{A}\|_F = 0.$$

In particular, if  $\{(\mathbf{A}_n, s_n)\}_{n=1}^\infty$  is a sequence in  $\Theta$  such that, for some  $(\mathbf{A}, s) \in \Theta$ , we have both  $s_n \rightarrow s$  and  $\Phi((\mathbf{A}_n, s_n)) \rightarrow \Phi((\mathbf{A}, s))$  as  $n \rightarrow \infty$ , then  $[\mathbf{A}_n, s_n] \rightarrow [\mathbf{A}, s]$  in  $M$  as  $n \rightarrow \infty$ .

*Proof.* Assume, for the sake of contradiction, that there exists an  $\epsilon > 0$  and a subsequence  $\{n_k\}_{k=1}^\infty$  such that, for all  $k$ ,

$$\min_{\mathbf{Q} \in O(q)} \|\mathbf{A}_{n_k} \mathbf{Q} - \mathbf{A}\|_F \geq \epsilon.$$

Given that  $\|\mathbf{A}_n\|_F^2 = \text{tr}(\mathbf{A}_n \mathbf{A}_n^T)$  is convergent, in particular bounded, choose subsubsequence  $\{n_{k_l}\}_{l=1}^\infty$  such that  $\mathbf{A}_{n_{k_l}} \rightarrow \mathbf{B}$  as  $l \rightarrow \infty$  for some  $p \times q$  matrix  $\mathbf{B}$ . It follows that  $\mathbf{B} \mathbf{B}^T = \mathbf{A} \mathbf{A}^T$ . Therefore,  $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{B}) = q$  and  $\mathbf{B} = \mathbf{A} \mathbf{P}$  for some  $\mathbf{P} \in O(q)$ . We have

$$\min_{\mathbf{Q} \in O(q)} \|\mathbf{A}_{n_{k_l}} \mathbf{Q} - \mathbf{A}\|_F \leq \|\mathbf{A}_{n_{k_l}} \mathbf{P}^T - \mathbf{A}\|_F = \|\mathbf{A}_{n_{k_l}} \mathbf{P}^T - \mathbf{B} \mathbf{P}^T\|_F = \|\mathbf{A}_{n_{k_l}} - \mathbf{B}\|_F \rightarrow 0$$

as  $l \rightarrow \infty$ , a contradiction.

Now suppose that  $\{(\mathbf{A}_n, s_n)\}_{n=1}^\infty$  is a sequence in  $\Theta$  such that, for some  $(\mathbf{A}, s) \in \Theta$ , we have both  $s_n \rightarrow s$  and  $\Phi((\mathbf{A}_n, s_n)) \rightarrow \Phi((\mathbf{A}, s))$  as  $n \rightarrow \infty$ . It follows that  $\mathbf{A}_n \mathbf{A}_n^T \rightarrow \mathbf{A} \mathbf{A}^T$  and the previous paragraph shows that there exists a sequence  $\{\mathbf{Q}_n\}_{n=1}^\infty$  of  $q \times q$  orthogonal matrices such that  $\mathbf{A}_n \mathbf{Q}_n \rightarrow \mathbf{A}$  as  $n \rightarrow \infty$ . Thus,

$$\lim_{n \rightarrow \infty} [\mathbf{A}_n, s_n] = \lim_{n \rightarrow \infty} \pi((\mathbf{A}_n \mathbf{Q}_n, s_n)) = \pi((\mathbf{A}, s)) = [\mathbf{A}, s].$$

The proof is complete.  $\square$

## 1.3 Properties of the Probabilistic Principal Component Analysis Model

We now show that the statistical model  $\mathcal{P}$  given by (1.7) is *differentiable in quadratic mean* and has *positive definite Fisher information operator* in the sense developed in Sun et al. (2026), definition III.1 and III.2.

Let us introduce a few notations. For each  $[\mathbf{W}, \sigma^2] \in M$ , let

$$p(\mathbf{x}; [\mathbf{W}, \sigma^2]) = (2\pi)^{-\frac{p}{2}} \det^{-\frac{1}{2}}(\Sigma([\mathbf{W}, \sigma^2])) \exp\left(-\frac{1}{2} \mathbf{x}^T (\Sigma([\mathbf{W}, \sigma^2]))^{-1} \mathbf{x}\right) \quad (1.30)$$

be the density of  $N_p(\mathbf{0}, \Sigma([\mathbf{W}, \sigma^2]))$ . For  $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}, \sigma^2]} \subset T_{[\mathbf{W}, \sigma^2]} M \cong \mathbb{R}^d$ , write  $l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$  for the log-likelihood function of the local experiment

$$l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) = \log\left(p(\mathbf{x}; \text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))\right). \quad (1.31)$$

Define the square root likelihood function  $r_{[\mathbf{W}, \sigma^2]} : \mathcal{E}_{[\mathbf{W}, \sigma^2]} \rightarrow L^2(\mathbb{R}^p)$  by setting, for all  $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}, \sigma^2]}$ , that

$$r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) = \sqrt{p(\mathbf{x}; \text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))}. \quad (1.32)$$

**Theorem 12.**  $r_{[\mathbf{W}, \sigma^2]}$  is Fréchet differentiable at any  $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}, \sigma^2]}$ . In particular,  $\mathcal{P}$  is differentiable in quadratic mean with score

$$s(\mathbf{x}; [\mathbf{W}, \sigma^2])(\mathbf{h}) = \frac{1}{2} \text{tr} \left\{ \left( \Sigma([\mathbf{W}, \sigma^2]) \right)^{-1} [\mathbf{x}\mathbf{x}^T - \left( \Sigma([\mathbf{W}, \sigma^2]) \right)] \left( \Sigma([\mathbf{W}, \sigma^2]) \right)^{-1} (D_{\mathbf{h}}\Sigma([\mathbf{W}, \sigma^2])) \right\}, \quad (1.33)$$

where

$$D_{\mathbf{h}}\Sigma([\mathbf{W}, \sigma^2]) := \left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(t\mathbf{h})) = \mathbf{U}\mathbf{W}^T + \mathbf{W}\mathbf{U}^T + b\mathbf{I}_p, \quad (1.34)$$

if  $\mathbf{h} \in T_{[\mathbf{W}, \sigma^2]}M$  lifts to  $(\mathbf{U}, b)$  at  $(\mathbf{W}, \sigma^2)$ . Furthermore,  $\mathcal{P}$  is a regular model in the sense that the Fisher information operator

$$G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}_1, \mathbf{h}_2) = \frac{1}{2} \text{tr} \left\{ \left( \Sigma([\mathbf{W}, \sigma^2]) \right)^{-1} D_{\mathbf{h}_1}\Sigma([\mathbf{W}, \sigma^2]) \left( \Sigma([\mathbf{W}, \sigma^2]) \right)^{-1} D_{\mathbf{h}_2}\Sigma([\mathbf{W}, \sigma^2]) \right\} \quad (1.35)$$

is positive definite.

*Proof.* The proof for the Fréchet differentiability is long but routine. We compute the Gâteaux derivative of  $r_{[\mathbf{W}, \sigma^2]}$  at  $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}, \sigma^2]}$  in a given direction  $\mathbf{h} \in T_{[\mathbf{W}, \sigma^2]}M$  and show that the Gâteaux derivative is the Fréchet derivative by a direct analysis of the remainder term. At the very end, we specialize our computations to  $\mathbf{v} = \mathbf{0}$  to prove (1.33) and (1.35).

Given  $\mathbf{v}$  and  $\mathbf{h}$ , there exists an  $\epsilon > 0$  such that  $\mathbf{v} + t\mathbf{h} \in \mathcal{E}_{[\mathbf{W}, \sigma^2]}$  whenever  $t \in (-\epsilon, \epsilon)$ . If they respectively lift to  $(\mathbf{V}, c)$  and  $(\mathbf{U}, b)$  at  $(\mathbf{W}, \sigma^2)$ , then  $\mathbf{v} + t\mathbf{h}$  lifts to  $(\mathbf{V} + t\mathbf{U}, c + tb)$  at  $(\mathbf{W}, \sigma^2)$ . First of all:

$$\begin{aligned} D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) &:= \left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})) \\ &= \left. \frac{d}{dt} \right|_{t=0} \left\{ (\mathbf{W} + \mathbf{V} + t\mathbf{U})(\mathbf{W} + \mathbf{V} + t\mathbf{U})^T + (\sigma^2 + c + tb)\mathbf{I}_p \right\} \\ &= \mathbf{U}\mathbf{V}^T + \mathbf{V}\mathbf{U}^T + \mathbf{U}\mathbf{W}^T + \mathbf{W}\mathbf{U}^T + b\mathbf{I}_p, \end{aligned} \quad (1.36)$$

proving (1.34) with special case  $\mathbf{v} = \mathbf{0}$ . Secondly, for all  $\mathbf{x} \in \mathbb{R}^p$ :

$$\begin{aligned} D_{\mathbf{h}}l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) &:= \left. \frac{d}{dt} \right|_{t=0} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) \\ &= \left. \frac{d}{dt} \right|_{t=0} \left\{ -\frac{p}{2} \log(2\pi) - \frac{1}{2} \log \det \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})) \right) - \frac{1}{2} \mathbf{x}^T \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})) \right)^{-1} \mathbf{x} \right\} \\ &= -\frac{1}{2} \text{tr} \left[ \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( \left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})) \right) \right] \\ &\quad + \frac{1}{2} \mathbf{x}^T \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( \left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})) \right) \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \mathbf{x} \\ &= -\frac{1}{2} \text{tr} \left[ \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \right] \\ &\quad + \frac{1}{2} \mathbf{x}^T \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \mathbf{x} \\ &= \frac{1}{2} \text{tr} \left\{ \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} [\mathbf{x}\mathbf{x}^T - \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)] \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \right\}. \end{aligned} \quad (1.37)$$

Finally:

$$\begin{aligned} D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) &= D_{\mathbf{h}} \exp \left( \frac{1}{2} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) \right) = \frac{1}{2} \exp \left( \frac{1}{2} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) \right) D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) \\ &= \frac{1}{2} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}). \end{aligned} \quad (1.38)$$

To prove that (1.38) indeed defines a valid Gâteaux derivative, we must prove that  $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}) \in L^2(\mathbb{R}^p)$ . Luckily, the expectation and covariance of normal quadratic forms have already been very well studied in [Gupta and Nagar \(2000\)](#). For  $\mathbf{X} \sim N_p(\mathbf{0}, \Sigma)$ , Theorem 7.7.1(i) in [Gupta and Nagar \(2000\)](#) states that

$$\mathbb{E} [\mathbf{X}^T \mathbf{Z} \mathbf{X}] = \text{tr}(\mathbf{Z} \Sigma)$$

for any  $p \times p$  matrix  $\mathbf{Z}$  and Theorem 7.7.2 in [Gupta and Nagar \(2000\)](#) states that

$$\text{Cov}(\mathbf{X}^T \mathbf{Z} \mathbf{X}, \mathbf{X}^T \mathbf{Y} \mathbf{X}) = \text{tr}(\mathbf{Z} \Sigma \mathbf{Y}^T \Sigma) + \text{tr}(\mathbf{Z}^T \Sigma \mathbf{Y}^T \Sigma)$$

for any two  $p \times p$  matrices  $\mathbf{Y}$  and  $\mathbf{Z}$ . Therefore, we can confirm without effort that the expectation of derivative of log density is zero:

$$\begin{aligned} &\int_{\mathbb{R}^p} D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))) d\mathbf{x} \\ &= -\frac{1}{2} \text{tr} \left[ \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \right] \\ &+ \frac{1}{2} \int_{\mathbb{R}^p} \left[ \mathbf{x}^T \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \mathbf{x} \right] p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))) d\mathbf{x} = 0 \end{aligned}$$

and, whenever  $\mathbf{h}_1$  and  $\mathbf{h}_2 \in T_{[\mathbf{W}, \sigma^2]}M$ :

$$\begin{aligned} &\int_{\mathbb{R}^p} D_{\mathbf{h}_1} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) D_{\mathbf{h}_2} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))) d\mathbf{x} \\ &= \frac{1}{4} \text{Cov}(\mathbf{X}^T \mathbf{Z}_1 \mathbf{X}, \mathbf{X}^T \mathbf{Z}_2 \mathbf{X}) = \frac{1}{4} \cdot 2 \text{tr} \left( \mathbf{Z}_1 \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \mathbf{Z}_2 \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \\ &= \frac{1}{2} \text{tr} \left\{ \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} D_{\mathbf{h}_1} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} D_{\mathbf{h}_2} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right\}, \end{aligned} \quad (1.39)$$

where  $\mathbf{X} \sim N_p(\mathbf{0}, \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})))$  and

$$\mathbf{Z}_j := \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}_j} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \quad (1.40)$$

is symmetric for  $j = 1, 2$ , so that the two terms in Theorem 7.7.2 of [Gupta and Nagar \(2000\)](#) coincide. Thus, in the particular case  $\mathbf{h} = \mathbf{h}_1 = \mathbf{h}_2$ :

$$\begin{aligned} \|D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\|^2 &= \int_{\mathbb{R}^p} [D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})]^2 d\mathbf{x} \\ &= \frac{1}{4} \int_{\mathbb{R}^p} (D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}))^2 p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))) d\mathbf{x} = \frac{1}{8} \text{tr} \left\{ \left[ \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right]^2 \right\}. \end{aligned} \quad (1.41)$$

Therefore,  $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}) \in L^2(\mathbb{R}^p)$  and (1.38) is indeed a valid Gâteaux derivative at  $\mathbf{v}$  in the direction  $\mathbf{h}$ .

To prove that this is indeed the Fréchet derivative, let us look at the remainder term:

$$R(\mathbf{v}, \mathbf{h})(\mathbf{x}) := r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + \mathbf{h})(\mathbf{x}) - r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) - D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})$$

and prove that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{1}{|\mathbf{h}|^2} \|R(\mathbf{v}, \mathbf{h}); L^2(\mathbb{R}^p)\|^2 = 0.$$

To start, the fundamental theorem of calculus shows that

$$\begin{aligned} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + \mathbf{h})(\mathbf{x}) - r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) &= \int_0^1 \frac{d}{dt} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) dt \\ &= \int_0^1 \lim_{s \rightarrow 0} \frac{r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + (t+s)\mathbf{h})(\mathbf{x}) - r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})(\mathbf{x})}{s} dt = \int_0^1 D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) dt. \end{aligned}$$

Therefore, by Minkowski's integral inequality and the linearity of  $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$  in  $\mathbf{h}$ , we have

$$\begin{aligned} \|R(\mathbf{v}, \mathbf{h}); L^2(\mathbb{R}^p)\| &= \left\| \int_0^1 (D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) - D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})) dt; L^2(\mathbb{R}^p) \right\| \\ &\leq \int_0^1 \|D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt = |\mathbf{h}| \int_0^1 \|D_{\frac{\mathbf{h}}{|\mathbf{h}|}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\frac{\mathbf{h}}{|\mathbf{h}|}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt \end{aligned}$$

and

$$\begin{aligned} \frac{1}{|\mathbf{h}|^2} \|R(\mathbf{v}, \mathbf{h}); L^2(\mathbb{R}^p)\|^2 &\leq \left\{ \int_0^1 \|D_{\frac{\mathbf{h}}{|\mathbf{h}|}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\frac{\mathbf{h}}{|\mathbf{h}|}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt \right\}^2 \\ &\leq \left\{ \int_0^1 \sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt \right\}^2. \end{aligned}$$

The proof will be complete once we can show that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \int_0^1 \sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt = 0. \quad (1.42)$$

We must therefore examine the continuity of  $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$ , which is easy to see if we revisit the formulas. (1.36) shows that  $D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))$  is smooth in both  $\mathbf{h}$  and  $\mathbf{v}$ . Thus, (1.38) shows that, for each fixed  $\mathbf{x} \in \mathbb{R}^p$ ,  $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})$  is smooth in both  $\mathbf{h}$  and  $\mathbf{v}$  and (1.41) shows that  $\|D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\|$  is smooth in both  $\mathbf{h}$  and  $\mathbf{v}$ . Scheffé's lemma shows that  $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$ , as a map to  $L^2(\mathbb{R}^p)$ , is continuous in  $\mathbf{h}$  and in  $\mathbf{v}$ . The integrand inside (1.42) is thus bounded uniformly over all  $t \in [0, 1]$  and all  $\mathbf{h}$  in a compact neighborhood of  $\mathbf{0}$  and it suffices to show that, for every  $t \in [0, 1]$ , that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| = 0, \quad (1.43)$$

thanks to the bounded convergence theorem. To prove (1.43), let  $\mathbf{e}_1, \dots, \mathbf{e}_d$  be a  $g_{[\mathbf{W}, \sigma^2]}$  orthonormal basis. Every unit vector  $\mathbf{u}$  can be expressed as  $\mathbf{u} = \sum_{i=1}^d u_i \mathbf{e}_i$  with  $\sum_{i=1}^d u_i^2 = 1$ . By linearity of  $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$  in  $\mathbf{h}$ , we have

$$\begin{aligned} \|D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| &\leq \sum_{i=1}^d |u_i| \|D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| \\ &\leq \sqrt{d} \max_{i=1, \dots, d} \|D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\|. \end{aligned}$$

Taking supremum yields

$$\sup_{\|\mathbf{u}\|=1} \|D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| \leq \sqrt{d} \max_{i=1, \dots, d} \|D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\|.$$

Given that  $D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) \rightarrow D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$  in  $L^2(\mathbb{R}^p)$  as  $\mathbf{h} \rightarrow \mathbf{0}$  for every  $i$ , every  $t$ , and every  $\mathbf{v}$ , the proof of Fréchet differentiability is complete.

Finally, the formulas (1.36), (1.37), and (1.38) simplifies when  $\mathbf{v} = \mathbf{0}$ :

$$\begin{aligned} D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{0})(\mathbf{x}) &= \frac{1}{2} r_{[\mathbf{W}, \sigma^2]}(\mathbf{0})(\mathbf{x}) D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{0})(\mathbf{x}) \\ &= \frac{1}{4} \sqrt{p(\mathbf{x}; [\mathbf{W}, \sigma^2])} \operatorname{tr} \left\{ (\Sigma([\mathbf{W}, \sigma^2]))^{-1} [\mathbf{x}\mathbf{x}^T - (\Sigma([\mathbf{W}, \sigma^2]))] (\Sigma([\mathbf{W}, \sigma^2]))^{-1} (D_{\mathbf{h}} \Sigma([\mathbf{W}, \sigma^2])) \right\}. \end{aligned}$$

The score  $s(\mathbf{x}; [\mathbf{W}, \sigma^2])(\mathbf{h})$  therefore equals to  $D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{0})(\mathbf{x})$  and is indeed given by (1.33). The Fisher information operator (1.35) is obtained by taking  $\mathbf{v} = \mathbf{0}$  in (1.39). To show that  $G_{[\mathbf{W}, \sigma^2]}$  is positive definite, let  $\mathbf{h} \in T_{[\mathbf{W}, \sigma^2]}M$  be given. We show that  $G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}, \mathbf{h}) \geq 0$  and equality holds only when  $\mathbf{h} = \mathbf{0}$ . The non-negative part is clear from (1.35):

$$G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}, \mathbf{h}) = \frac{1}{2} \operatorname{tr} \left\{ \left[ (\Sigma([\mathbf{W}, \sigma^2]))^{-1} D_{\mathbf{h}} \Sigma([\mathbf{W}, \sigma^2]) \right]^2 \right\} \geq 0.$$

Assume from now on that the equality holds. A slight annoyance here is that the product of two symmetric matrices is not necessarily symmetric and we cannot immediately conclude that  $D_{\mathbf{h}} \Sigma([\mathbf{W}, \sigma^2]) = \mathbf{0}$ . We easily overcome this difficulty by using the cyclic property of the trace to get a symmetric matrix

$$\mathbf{N} := (\Sigma([\mathbf{W}, \sigma^2]))^{-\frac{1}{2}} D_{\mathbf{h}} \Sigma([\mathbf{W}, \sigma^2]) (\Sigma([\mathbf{W}, \sigma^2]))^{-\frac{1}{2}}$$

such that

$$0 = G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}, \mathbf{h}) = \frac{1}{2} \operatorname{tr}(\mathbf{N}^2).$$

Therefore,  $\mathbf{N} = \mathbf{0}$  and  $D_{\mathbf{h}} \Sigma([\mathbf{W}, \sigma^2]) = \mathbf{0}$ . Now (1.34) reads the same as (1.9) in the proof of Proposition 1. We therefore get  $(\mathbf{U}, b) \in \ker(d\Phi_{(\mathbf{W}, \sigma^2)}) = V_{(\mathbf{W}, \sigma^2)}$ . However, being lifted from  $\mathbf{h}$ ,  $(\mathbf{U}, b)$  must also be a horizontal vector. Therefore,  $\mathbf{h} = \mathbf{0}$ . We have proven that  $G_{[\mathbf{W}, \sigma^2]}$  is positive definite.  $\square$

As an immediate consequence, we have local asymptotic normality.

**Proposition 13.** With the true parameter  $[\theta_0] \in M$ , let us write

$$[\theta_{n, \mathbf{h}}] = \operatorname{Exp}_{[\theta_0]} \left( \frac{\mathbf{h}}{\sqrt{n}} \right), \quad (1.44)$$

whenever  $n$  is large enough such that  $\frac{\mathbf{h}}{\sqrt{n}} \in U_{[\theta_0]}$ . For any  $[\theta] \in M$ , let us write  $P_{[\theta]}^n$  for the joint distribution of  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  when all components are independent and identically distributed as  $N_p(\mathbf{0}, \Sigma([\theta]))$ .

We have the local asymptotic normality expansion

$$\sum_{i=1}^n \log \frac{p(\mathbf{X}_i; [\theta_{n, \mathbf{h}}])}{p(\mathbf{X}_i; [\theta_0])} - \sum_{i=1}^n s(\mathbf{X}_i; [\theta_0]) \left( \frac{\mathbf{h}}{\sqrt{n}} \right) + \frac{1}{2} G_{[\theta_0]}(\mathbf{h}, \mathbf{h}) \rightarrow 0, \quad (1.45)$$

in probability as  $n \rightarrow \infty$  when  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  follows  $P_{\theta_0}^n$ .

In particular,

$$\sum_{i=1}^n \log \frac{p(\mathbf{X}_i; [\theta_{n, \mathbf{h}}])}{p(\mathbf{X}_i; [\theta_0])} \rightarrow N \left( -\frac{1}{2} G_{[\theta_0]}(\mathbf{h}, \mathbf{h}), G_{[\theta_0]}(\mathbf{h}, \mathbf{h}) \right) \quad (1.46)$$

in distribution as  $n \rightarrow \infty$  when  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  follows  $P_{[\theta_0]}^n$  and Le Cam's first lemma in the format of Example 6.5 in van der Vaart (1998) shows that  $P_{[\theta_0]}^n$  and  $P_{[\theta_{n, \mathbf{h}}]}^n$  are mutually contiguous.

*Proof.* Given that  $\mathcal{P}$  is differentiable in quadratic mean and has positive definite Fisher operator, this follows from Proposition III.1 in [Sun et al. \(2026\)](#).  $\square$

## 1.4 Properties of the Maximum Likelihood Estimator of $\mathcal{P}$

### 1.4.1 The Maximum Likelihood Estimator

In this section, we assume that there is a true parameter  $[\mathbf{W}_0, \sigma_0^2] \in M$  for the model  $\mathcal{P}$  given in (1.7) and we investigate the consistency and asymptotic normality of the maximum likelihood estimator towards this true parameter.

More concretely, given data points  $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$ , we consider them as realizations of independent identically distributed random variables  $\mathbf{X}_1, \dots, \mathbf{X}_n$  whose true density is  $p(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])$ , where  $p$  is given in (1.30). In what follows, we denote the true covariance matrix by

$$\Sigma_0 := \Sigma([\mathbf{W}_0, \sigma_0^2]) = \mathbf{W}_0 \mathbf{W}_0^T + \sigma_0^2 \mathbf{I}_p. \quad (1.47)$$

The eigenvalues of the true covariance matrix are

$$\lambda_j(\Sigma_0) = \begin{cases} \sigma_j(\mathbf{W}_0)^2 + \sigma_0^2 & j = 1, \dots, q \\ \sigma_0^2 & j = q+1, \dots, p. \end{cases} \quad (1.48)$$

The sample covariance matrix is defined by

$$\mathbf{S}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^T. \quad (1.49)$$

Given that we will discuss  $\mathbf{S}_n$  a lot, we shall write  $\lambda_j$  in place of  $\lambda_j(\mathbf{S}_n)$ . We need also to fix one choice of orthonormal basis of eigenvectors  $\mathbf{v}_1, \dots, \mathbf{v}_p$  where  $\mathbf{v}_j = \mathbf{v}_j(\mathbf{S}_n)$  corresponds to the eigenvalue  $\lambda_j$ . The facts we will use repeatedly is that, when  $p$  and  $q$  are both fixed,  $\mathbf{S}_n \rightarrow \Sigma_0$  almost surely as  $n \rightarrow \infty$  and, due to the fact that eigenvalues depends continuously on the matrices,  $\lambda_j \rightarrow \lambda_j(\Sigma_0)$  almost surely as  $n \rightarrow \infty$ .

We now formulate the maximum likelihood estimator as computed in [Tipping and Bishop \(1999\)](#). Put  $\mathbf{U} = [\mathbf{v}_1, \dots, \mathbf{v}_q]$ , a  $p \times q$  matrix, and  $\Delta_q$  to be a  $q \times q$  diagonal matrix with diagonal entries  $\lambda_1, \dots, \lambda_q$ . Then, the collection of all choices of  $(\hat{\mathbf{W}}_n, \hat{\sigma}_n^2)$  maximizing the likelihood is given by

$$\hat{\mathbf{W}}_n = \mathbf{U} (\Delta_q - \hat{\sigma}_n^2 \mathbf{I}_q)^{\frac{1}{2}} \mathbf{R} \quad (1.50)$$

and

$$\hat{\sigma}_n^2 = \frac{1}{p-q} \sum_{j=q+1}^p \lambda_j, \quad (1.51)$$

ranged over all rotation matrices  $\mathbf{R} \in O(q)$ .

It is immediately clear from (1.48) that, almost surely, for all sufficiently large  $n$ ,  $\lambda_j > 0$  for all  $j$  and  $\hat{\sigma}_n^2 > 0$  so that  $\hat{\sigma}_n^2$  is well-defined. Also,  $\hat{\sigma}_n^2 \rightarrow \sigma_0^2$  almost surely as  $n \rightarrow \infty$ . Another look at (1.48) shows that, almost surely, for all sufficiently large  $n$ ,  $\lambda_j - \hat{\sigma}_n^2 > 0$  for all  $j \in \{1, \dots, q\}$ ,  $\lambda_q > \lambda_{q+1}$ , and  $\text{rank}(\hat{\mathbf{W}}_n) = q$  so that  $\hat{\mathbf{W}}_n$  is well-defined. Thus, almost surely, for all sufficiently large  $n$ , the same model parametrized by  $M$  has a unique maximum likelihood estimator  $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]$ .

### 1.4.2 Consistency

First of all, I would like to argue that this quotient manifold parameter space allows us to settle consistency in an (arguably more straightforward) way that does not use [Datta and Chakrabarty \(2023\)](#) or [Redner \(1981\)](#).

**Theorem 14.** *The maximum likelihood estimator is consistent. That is,  $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \rightarrow [\mathbf{W}_0, \sigma_0^2]$  almost surely as  $n \rightarrow \infty$ .*

*Proof.* Put

$$\hat{\Sigma}_n := \Sigma \left( [\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \right) = \hat{\mathbf{W}}_n \hat{\mathbf{W}}_n^T + \hat{\sigma}_n^2 \mathbf{I}_p. \quad (1.52)$$

In light of Proposition 11 and the fact that  $\hat{\sigma}_n^2$  is consistent, it suffices to prove that  $\hat{\Sigma}_n \rightarrow \Sigma_0$  almost surely as  $n \rightarrow \infty$ . The fact that  $\sum_{j=1}^p \mathbf{v}_j \mathbf{v}_j^T = \mathbf{I}_p$  allows us to write

$$\hat{\Sigma}_n = \mathbf{U} (\Delta_q - \hat{\sigma}_n^2 \mathbf{I}_q) \mathbf{U}^T + \hat{\sigma}_n^2 \mathbf{I}_p = \sum_{j=1}^q (\lambda_j - \hat{\sigma}_n^2) \mathbf{v}_j \mathbf{v}_j^T + \hat{\sigma}_n^2 \mathbf{I}_p = \sum_{j=1}^q \lambda_j \mathbf{v}_j \mathbf{v}_j^T + \hat{\sigma}_n^2 \sum_{j=q+1}^p \mathbf{v}_j \mathbf{v}_j^T$$

and

$$\mathbf{S}_n = \mathbf{S}_n \sum_{j=1}^p \mathbf{v}_j \mathbf{v}_j^T = \sum_{j=1}^p \lambda_j \mathbf{v}_j \mathbf{v}_j^T.$$

Thus,

$$\hat{\Sigma}_n - \mathbf{S}_n = \sum_{j=q+1}^p (\hat{\sigma}_n^2 - \lambda_j) \mathbf{v}_j \mathbf{v}_j^T$$

and

$$\|\hat{\Sigma}_n - \mathbf{S}_n\| \leq \sum_{j=q+1}^p |\hat{\sigma}_n^2 - \lambda_j| \|\mathbf{v}_j \mathbf{v}_j^T\| = \sum_{j=q+1}^p |\hat{\sigma}_n^2 - \lambda_j|.$$

Therefore, given that both  $\hat{\sigma}_n^2$  and  $\lambda_j$  will converge almost surely to  $\sigma_0^2$  when  $j \in \{q+1, \dots, p\}$ ,  $\|\hat{\Sigma}_n - \mathbf{S}_n\| \rightarrow 0$  almost surely as  $n \rightarrow \infty$ . The proof is complete.  $\square$

### 1.4.3 Asymptotic Linearity

We now move to prove that, locally at the true parameter, the maximum likelihood estimator is asymptotically normal in the tangent space of the true parameter by local linearity. Given that the manifold is locally Euclidean, this result can be attacked with the classical approach, namely by verifying conditions of Theorem 5.23 in [van der Vaart \(1998\)](#), without too much use of geometric analysis. We start with computational lemmas.

**Lemma 15.** Given  $[\mathbf{W}, \sigma^2] \in M$  and any convex neighborhood  $U$  of  $\mathbf{0}$  in  $T_{[\mathbf{W}, \sigma^2]} M$  such that  $\bar{U} \subset \mathcal{E}_{[\mathbf{W}, \sigma^2]}$ , there exists a function  $m : \mathbb{R}^p \rightarrow \mathbb{R}$  such that, whenever  $\mathbf{v}_1, \mathbf{v}_2 \in U$  and  $\mathbf{x} \in \mathbb{R}^p$ , we always have

$$|l_{[\mathbf{W}, \sigma^2]}(\mathbf{v}_1)(\mathbf{x}) - l_{[\mathbf{W}, \sigma^2]}(\mathbf{v}_2)(\mathbf{x})| \leq m(\mathbf{x}) |\mathbf{v}_1 - \mathbf{v}_2|.$$

This  $m$  has its second moment under any distribution  $\mu$  on  $\mathbb{R}^p$  that has the fourth moment. That is,

$$\int_{\mathbb{R}^p} (m(\mathbf{x}))^2 d\mu(\mathbf{x}) < \infty$$

whenever

$$\int_{\mathbb{R}^p} |\mathbf{x}|^4 d\mu(\mathbf{x}) < \infty.$$

*Proof.* Given  $[\mathbf{W}, \sigma^2] \in M$  and a convex neighborhood  $U$  of  $\mathbf{0}$  compactly contained in  $\mathcal{E}_{[\mathbf{W}, \sigma^2]}$ , the result will follow by the mean value inequality if we can find a function  $m$  such that

$$|D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})| \leq m(\mathbf{x})$$

for all unit vector  $\mathbf{h} \in T_{[\mathbf{W}, \sigma^2]}M$ , all  $\mathbf{v} \in \bar{U}$ , and all  $\mathbf{x} \in \mathbb{R}^p$ . Given that we have proved in Proposition 12 that  $\left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))\right)^{-1}$  is continuous in  $\mathbf{v}$  and that  $D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))$  is continuous in both  $\mathbf{h}$  and  $\mathbf{v}$ , we choose constants  $C_1 \in (0, \infty)$  and  $C_2 \in (0, \infty)$  such that

$$\frac{1}{2} \left\| \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \right\|_F \leq C_1$$

and

$$\frac{1}{2} \left\| \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \right\|_F \leq C_2$$

for all all unit vector  $\mathbf{h} \in T_{[\mathbf{W}, \sigma^2]}M$  and all  $\mathbf{v} \in \bar{U}$ , thanks to the fact that  $\bar{U}$  is compact. from Proposition 8. By (1.37), we have

$$\begin{aligned} |D_{\mathbf{h}}l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})| &\leq \frac{1}{2} \|\mathbf{x}\mathbf{x}^T\|_F \left\| \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \right\|_F \\ &\quad + \frac{1}{2} \left\| \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \right\|_F \leq C_1 \mathbf{x}^T \mathbf{x} + C_2 \end{aligned}$$

Setting  $m(\mathbf{x}) = C_1 \mathbf{x}^T \mathbf{x} + C_2$  will finish the proof.  $\square$

**Lemma 16.** Define function  $L : \mathcal{E}_{[\mathbf{W}_0, \sigma_0^2]} \rightarrow \mathbb{R}$  by setting, for all  $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}_0, \sigma_0^2]}$ ,

$$L(\mathbf{v}) = \int_{\mathbb{R}^p} l_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v})(\mathbf{x}) p(\mathbf{x}; \Sigma_0) d\mathbf{x}.$$

Then  $\mathbf{0}$  is the unique maximizer of  $L$  and, for all  $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}_0, \sigma_0^2]}$ :

$$L(\mathbf{v}) = L(\mathbf{0}) - \frac{1}{2} G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v}, \mathbf{v}) + R(\mathbf{v})$$

with the remainder of the second order Taylor expansion satisfying

$$\lim_{|\mathbf{v}| \rightarrow 0} \frac{R(\mathbf{v})}{|\mathbf{v}|^2} = 0.$$

*Proof.* One should immediately recognize that  $L(\mathbf{0}) - L(\mathbf{v})$  equals to the Kullback-Leibler divergence of the true model  $N_p(\mathbf{0}, \Sigma_0)$  with respect to the model  $N_p\left(\mathbf{0}, \Sigma\left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v})\right)\right)$ . Therefore,  $L(\mathbf{v})$  is maximized at all  $\mathbf{v}$  with  $\Sigma\left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v})\right) = \Sigma_0$ . Given that  $\Sigma$  is injective and that we have proven in Proposition 8 that  $\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}$  is injective in  $\mathcal{E}_{[\mathbf{W}_0, \sigma_0^2]}$ , this only happens when  $\mathbf{v} = \mathbf{0}$  and  $\mathbf{0}$  is the unique maximizer of  $L$ .

To obtain the Taylor expansion, we basically have to repeat the computation part in the proof of Proposition 12 on a slightly different problem with the second derivative at  $\mathbf{0}$  being computed as well. The first order derivative at  $\mathbf{0}$  is automatically 0 because  $\mathbf{0}$  is the maximizer.

Let  $\mathbf{h} \in T_{[\mathbf{W}_0, \sigma_0^2]}M$  be given. We can easily compute that

$$L(t\mathbf{h}) = -\frac{p}{2} \log(2\pi) - \frac{1}{2} \log \left( \det \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right) - \frac{1}{2} \text{tr} \left\{ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\}.$$

Suppose that  $\mathbf{h}$  lifts to  $(U, b)$  at  $(\mathbf{W}_0, \sigma_0^2)$ . (1.36) immediately shows that

$$\left. \frac{d^2}{dt^2} \right|_{t=0} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) = 2UU^T.$$

We now compute the second derivative  $L$  term by term. For the log determinant term, we have

$$\begin{aligned} \frac{d}{dt} \left( -\frac{1}{2} \log \left( \det \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right) \right) &= -\frac{1}{2} \text{tr} \left[ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d}{dt} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right], \\ \frac{d^2}{dt^2} \left( -\frac{1}{2} \log \left( \det \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right) \right) &= -\frac{1}{2} \frac{d}{dt} \text{tr} \left[ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d}{dt} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right] \\ &= -\frac{1}{2} \text{tr} \left\{ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d^2}{dt^2} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right\} \\ &\quad + \frac{1}{2} \text{tr} \left\{ \left[ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d}{dt} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right]^2 \right\}, \end{aligned}$$

and therefore

$$\frac{d^2}{dt^2} \Big|_{t=0} \left( -\frac{1}{2} \log \left( \det \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right) \right) = -\text{tr} \{ \Sigma_0^{-1} \mathbf{U} \mathbf{U}^T \} + \frac{1}{2} \text{tr} \{ [\Sigma_0^{-1} D_{\mathbf{h}} \Sigma([\mathbf{W}_0, \sigma_0^2])]^2 \}.$$

Next, for the trace term, we have

$$\begin{aligned} &\frac{d}{dt} \left( -\frac{1}{2} \text{tr} \left\{ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\} \right) \\ &= \frac{1}{2} \text{tr} \left\{ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d}{dt} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\}, \\ &\frac{d^2}{dt^2} \left( -\frac{1}{2} \text{tr} \left\{ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\} \right) \\ &= \frac{1}{2} \text{tr} \left\{ -2 \left\{ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d}{dt} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \right. \right. \\ &\quad \left. \frac{d}{dt} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\} \\ &\quad \left. + \frac{1}{2} \text{tr} \left\{ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d^2}{dt^2} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\}, \end{aligned}$$

and therefore

$$\frac{d^2}{dt^2} \Big|_{t=0} \left( -\frac{1}{2} \text{tr} \left\{ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\} \right) = -\text{tr} \{ [\Sigma_0^{-1} D_{\mathbf{h}} \Sigma([\mathbf{W}_0, \sigma_0^2])]^2 \} + \text{tr} (\Sigma_0^{-1} \mathbf{U} \mathbf{U}^T).$$

We end up the negative of the Fisher information operator, as computed in (1.35).

$$\frac{d^2}{dt^2} \Big|_{t=0} L(t\mathbf{h}) = -\frac{1}{2} \text{tr} \left\{ [\Sigma_0^{-1} D_{\mathbf{h}} \Sigma([\mathbf{W}_0, \sigma_0^2])]^2 \right\} = -G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{h}, \mathbf{h}).$$

This completes the proof.  $\square$

**Definition 17.** We define the **inverse Fisher information operator**  $G_{[\mathbf{W}, \sigma^2]}^{-1}$  as the inverse of the flat version of the Fisher information bilinear form. More concretely, instead of treating  $G_{[\mathbf{W}, \sigma^2]}$  as a bilinear form on  $T_{[\mathbf{W}, \sigma^2]}M$ , we treat it as a map  $G_{[\mathbf{W}, \sigma^2]} : T_{[\mathbf{W}, \sigma^2]}M \rightarrow T_{[\mathbf{W}, \sigma^2]}^*M$  by setting  $G_{[\mathbf{W}, \sigma^2]}(\mathbf{v}) = G_{[\mathbf{W}, \sigma^2]}(\mathbf{v}, \cdot)$  and we define  $G_{[\mathbf{W}, \sigma^2]}^{-1} : T_{[\mathbf{W}, \sigma^2]}^*M \rightarrow T_{[\mathbf{W}, \sigma^2]}M$  to be the inverse.

**Theorem 18.** *The maximum likelihood estimator is asymptotically linear at  $[\mathbf{W}_0, \sigma_0^2]$  with the influence function  $IF : \mathbb{R}^p \rightarrow T_{[\mathbf{W}_0, \sigma_0^2]}M$  given by*

$$IF(\mathbf{x}) = G_{[\mathbf{W}_0, \sigma_0^2]}^{-1}(s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])). \quad (1.53)$$

As described in [Sun et al. \(2026\)](#) Lemma III.3, this means

$$\sqrt{n} \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1} \left( [\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \right) - \frac{1}{\sqrt{n}} \sum_{i=1}^n IF(\mathbf{X}_i) \rightarrow 0 \quad (1.54)$$

in probability as  $n \rightarrow \infty$  when  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  follows the true distribution  $P_{[\mathbf{W}_0, \sigma_0^2]}^n$ . Also,  $IF(\mathbf{X})$  has mean  $\mathbf{0}$  and is square integrable if  $\mathbf{X}$  follows the true distribution  $N_p(\mathbf{0}, \Sigma_0)$ .

In particular, once we fix a  $g_{[\mathbf{W}_0, \sigma_0^2]}$  orthonormal basis  $\mathbf{e}_1, \dots, \mathbf{e}_d$  and we identify  $T_{[\mathbf{W}_0, \sigma_0^2]}M$  canonically as  $\mathbb{R}^d$  (in concrete terms, identify  $\sum_{j=1}^d a_j \mathbf{e}_j$  to the point  $(a_1, \dots, a_d) \in \mathbb{R}^d$ ), then  $\sqrt{n} \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1} \left( [\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \right)$  converges in distribution to  $N_d(\mathbf{0}, \mathbf{G}^{-1})$  as  $n \rightarrow \infty$  when  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  follows the true distribution  $P_{[\mathbf{W}_0, \sigma_0^2]}^n$ , where  $\mathbf{G}$  is the  $d \times d$  matrix defined by  $\mathbf{G}_{ij} = G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{e}_i, \mathbf{e}_j)$ .

*Proof.* Fix a  $g_{[\mathbf{W}_0, \sigma_0^2]}$  orthonormal basis  $\mathbf{e}_1, \dots, \mathbf{e}_d$  of  $T_{[\mathbf{W}_0, \sigma_0^2]}M$ . Define the matrix  $\mathbf{G} \in \mathbb{R}^{d \times d}$  as in the statement of lemma and the vector  $\nabla l(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2]) \in \mathbb{R}^d$  by setting  $(\nabla l(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2]))_j = s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])(\mathbf{e}_j)$ , for all  $j = 1, \dots, d$ . Identifying  $\sqrt{n} \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1} \left( [\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \right) \in T_{[\mathbf{W}_0, \sigma_0^2]}M$  to a point in  $\mathbb{R}^d$  through this basis, we immediately have, by combining the two previous lemmas and Theorem 5.23 in [van der Vaart \(1998\)](#), that

$$\sqrt{n} \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1} \left( [\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \right) - \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{G}^{-1} \nabla l(\mathbf{X}_i; [\mathbf{W}_0, \sigma_0^2]) \rightarrow 0 \quad (1.55)$$

in probability as  $n \rightarrow \infty$ . In particular,  $\sqrt{n} \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1} \left( [\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \right)$  converges in distribution to  $N_d(\mathbf{0}, \mathbf{G}^{-1})$  as  $n \rightarrow \infty$ . Also, if  $\mathbf{X}$  follows the true distribution  $N_p(\mathbf{0}, \Sigma_0)$ , then  $\mathbb{E}[\nabla l(\mathbf{X}; [\mathbf{W}_0, \sigma_0^2])] = \mathbf{0}$  and thus  $\mathbb{E}[\mathbf{G}^{-1} \nabla l(\mathbf{X}; [\mathbf{W}_0, \sigma_0^2])] = \mathbf{0}$ . Given that  $\nabla l(\mathbf{X}; [\mathbf{W}_0, \sigma_0^2])$  has its second moment, so does  $\mathbf{G}^{-1} \nabla l(\mathbf{X}; [\mathbf{W}_0, \sigma_0^2])$ .

We remark that (1.54) is simply (1.55) rephrased in a way using only vectors in  $T_{[\mathbf{W}_0, \sigma_0^2]}M$  defined without  $\mathbf{e}_1, \dots, \mathbf{e}_d$  and hence without the need of identifying  $T_{[\mathbf{W}_0, \sigma_0^2]}M$  with  $\mathbb{R}^d$ . Let us for simplicity define  $\mathbf{y} := \mathbf{G}^{-1} \nabla l(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2]) \in \mathbb{R}^d$ . Put  $\mathbf{v} \in T_{[\mathbf{W}_0, \sigma_0^2]}M$  by setting  $\mathbf{v} = \sum_{j=1}^d \mathbf{y}_j \mathbf{e}_j \in T_{[\mathbf{W}_0, \sigma_0^2]}M$ . We claim that  $\mathbf{v}$  is the unique vector such that, for all  $\mathbf{z} \in T_{[\mathbf{W}_0, \sigma_0^2]}M$ :

$$G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v}, \mathbf{z}) = s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])(\mathbf{z}). \quad (1.56)$$

The uniqueness of  $\mathbf{v}$  is immediate by the positive definiteness of  $G_{[\mathbf{W}_0, \sigma_0^2]}$ . Given that  $(\mathbf{G}\mathbf{y})_j = (\nabla l(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2]))_j$  for all  $j = 1, \dots, d$ , we have, for all  $j = 1, \dots, d$ :

$$G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v}, \mathbf{e}_j) = \sum_{i=1}^d \mathbf{y}_i G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{e}_i, \mathbf{e}_j) = \sum_{i=1}^d \mathbf{y}_i \mathbf{G}_{ij} = s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])(\mathbf{e}_j).$$

The claim is therefore proved. By looking at Definition 17, (1.56) reads  $G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v}) = s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])$  and hence  $\mathbf{v} = G_{[\mathbf{W}_0, \sigma_0^2]}^{-1}(s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2]))$ , giving (1.54). The results about the integral of  $IF(\mathbf{X})$  translates to here as well. This completes the proof.  $\square$

**Theorem 19.** *We have that*

$$\sqrt{n} \left( \hat{\Sigma}_n - \Sigma_0 \right) - \frac{1}{\sqrt{n}} \sum_{i=1}^n D_{IF(\mathbf{X}_i)} \Sigma([\mathbf{W}_0, \sigma_0^2]) \rightarrow \mathbf{0} \quad (1.57)$$

in probability as  $n \rightarrow \infty$  when  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  follows the true distribution  $P_{[\mathbf{W}_0, \sigma_0^2]}^n$ .

As in Theorem 18, let  $g_{[\mathbf{W}_0, \sigma_0^2]}$ -orthonormal basis  $\mathbf{e}_1, \dots, \mathbf{e}_d$  be fixed and  $\mathbf{G}$  be the  $d \times d$  matrix corresponding to  $G_{[\mathbf{W}_0, \sigma_0^2]}$ . Define a function  $D : \mathbb{R}^d \rightarrow \mathbb{R}^{p \times p}$  by setting, for all  $\mathbf{z} = (z_1, \dots, z_d)$ :

$$D(\mathbf{z}) = D_{\sum_{i=1}^d z_i \mathbf{e}_i} \Sigma([\mathbf{W}_0, \sigma_0^2]).$$

Then,

$$\sqrt{n} \left( \hat{\Sigma}_n - \Sigma_0 \right) \rightarrow D(\mathbf{Z})$$

in distribution as  $n \rightarrow \infty$  when  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  follows the true distribution  $P_{[\mathbf{W}_0, \sigma_0^2]}^n$ , where  $\mathbf{Z}$  is a random variable following  $N_d(\mathbf{0}, \mathbf{G}^{-1})$  distribution.

*Proof.* Suppose that a given tangent vector  $\mathbf{h} \in \mathcal{E}_{[\mathbf{W}_0, \sigma_0^2]}$  lifts horizontally to  $(\mathbf{U}, b)$  at  $(\mathbf{W}_0, \sigma_0^2)$ . (1.36) shows that

$$\Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{h}) \right) = (\mathbf{W}_0 + \mathbf{U})(\mathbf{W}_0 + \mathbf{U})^T + (\sigma_0^2 + b)\mathbf{I}_p = \Sigma_0 + \mathbf{U}\mathbf{U}^T + D_{\mathbf{h}}\Sigma([\mathbf{W}_0, \sigma_0^2]).$$

For simplicity, put

$$\mathbf{h}_n := \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1} \left( [\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \right).$$

If  $\mathbf{h}_n$  lifts horizontally to  $(\mathbf{U}_n, b_n)$  at  $(\mathbf{W}_0, \sigma_0^2)$ , then

$$\sqrt{n} \left( \hat{\Sigma}_n - \Sigma_0 \right) = \sqrt{n} \mathbf{U}_n \mathbf{U}_n^T + \sqrt{n} D_{\mathbf{h}_n} \Sigma([\mathbf{W}_0, \sigma_0^2]).$$

Since  $|\mathbf{U}_n \mathbf{U}_n^T|^2 = \|\mathbf{U}_n\|_F^2 = |\mathbf{h}_n|^2$ , the asymptotic normality of  $\sqrt{n} \mathbf{h}_n$  proved in Theorem 18 implies that  $\sqrt{n} \mathbf{U}_n \mathbf{U}_n^T \rightarrow \mathbf{0}$  in probability as  $n \rightarrow \infty$  when  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  follows the true distribution  $P_{[\mathbf{W}_0, \sigma_0^2]}^n$ . As of the other term, given that the directional derivative is a linear map with respect to  $\mathbf{h}$ , it follows from (1.54) that

$$\sqrt{n} \mathbf{h}_n - \frac{1}{\sqrt{n}} \sum_{i=1}^n D_{\text{IF}(\mathbf{X}_i)} \Sigma([\mathbf{W}_0, \sigma_0^2]) \rightarrow 0$$

in probability as  $n \rightarrow \infty$  when  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  follows the true distribution  $P_{[\mathbf{W}_0, \sigma_0^2]}^n$ . We have therefore proved (1.57). The rest of the conclusions follows immediately.  $\square$

#### 1.4.4 Local Regularity

We now investigate the behaviour of the maximum likelihood estimator when the underlying data has a distribution slightly perturbed from the true parameter and eventually show that the maximum likelihood estimator is locally regular in the sense described in Sun et al. (2026). Given that calculations regarding components of a point in  $M$  does not matter that much in this section, we shall proceed with the notations already introduced in Proposition 13.

In concrete terms, let  $[\hat{\theta}_n] = [\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]$  be the maximum likelihood estimator and, for any  $\mathbf{u} \in \mathcal{E}_{[\theta_0]}$ , let

$$\Gamma_{\mathbf{u}} := \Pi_{\text{Exp}_{[\theta_0]}(\mathbf{u})}^{[\theta_0]} : T_{\text{Exp}_{[\theta_0]}(\mathbf{u})} M \rightarrow T_{[\theta_0]} M$$

be the parallel transport map considered in Proposition 10. Given  $\mathbf{h} \in T_{[\theta_0]} M$ , our goal of the subsection is to show the asymptotic normality of

$$\sqrt{n} \Gamma_{\frac{\mathbf{h}}{\sqrt{n}}} \left( \text{Exp}_{[\theta_n, \mathbf{h}]}^{-1} \left( [\hat{\theta}_n] \right) \right)$$

as  $n \rightarrow \infty$  when  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  follows  $P_{[\theta_n, \mathbf{h}]}^n$ . We start with technical intermediate results.

**Lemma 20.** Suppose that  $K$  is a compact convex subset of  $\mathbb{R}^d \times \mathbb{R}^d$ ,  $R : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^d$  is a smooth function,  $R(\mathbf{u}, \mathbf{u}) = \mathbf{0}$  for every  $\mathbf{u} \in \mathbb{R}^d$ , and  $R(\mathbf{0}, \mathbf{w}) = \mathbf{0}$  for every  $\mathbf{w} \in \mathbb{R}^d$ . Then, there exists a constant  $C \in (0, \infty)$  such that, for all  $(\mathbf{u}, \mathbf{w}) \in K$ :

$$|R(\mathbf{u}, \mathbf{w})| \leq C \|\mathbf{u}\| \|\mathbf{w} - \mathbf{u}\|.$$

*Proof.* By the fundamental theorem of calculus, we have

$$\begin{aligned} R(\mathbf{u}, \mathbf{w}) &= \int_0^1 \frac{\partial R}{\partial \mathbf{w}}(\mathbf{u}, \mathbf{u} + t(\mathbf{w} - \mathbf{u}))(\mathbf{w} - \mathbf{u}) dt \\ &= \int_0^1 \int_0^1 \frac{d}{ds} \left[ \frac{\partial R}{\partial \mathbf{w}}(s\mathbf{u}, \mathbf{u} + t(\mathbf{w} - \mathbf{u}))(\mathbf{w} - \mathbf{u}) \right] ds dt = \int_0^1 \int_0^1 \frac{\partial}{\partial \mathbf{u}} \left[ \frac{\partial R}{\partial \mathbf{w}}(s\mathbf{u}, \mathbf{u} + t(\mathbf{w} - \mathbf{u}))(\mathbf{w} - \mathbf{u}) \right] \mathbf{u} ds dt, \end{aligned}$$

where we have used the fact that  $\frac{\partial R}{\partial \mathbf{w}} R(\mathbf{0}, \mathbf{u} + t(\mathbf{w} - \mathbf{u})) \equiv \mathbf{0}$  because  $R(\mathbf{0}, \mathbf{u} + t(\mathbf{w} - \mathbf{u})) \equiv \mathbf{0}$ . With

$$C := \max_{(\mathbf{u}, \mathbf{w}) \in K} \left\| \frac{\partial}{\partial \mathbf{u}} \frac{\partial R}{\partial \mathbf{w}}(\mathbf{u}, \mathbf{w}) \right\|,$$

which is finite because  $K$  is compact, we have, for all  $(\mathbf{u}, \mathbf{w}) \in K$ :

$$|R(\mathbf{u}, \mathbf{w})| \leq C \int_0^1 \int_0^1 \|\mathbf{u}\| \|\mathbf{w} - \mathbf{u}\| ds dt = C \|\mathbf{u}\| \|\mathbf{w} - \mathbf{u}\|.$$

The proof is complete.  $\square$

**Lemma 21.** Let  $T_n = T_n(\mathbf{X}_1, \dots, \mathbf{X}_n) \in M$  be an estimator such that, when  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  follows  $P_{[\theta_0]}^n$ , the sequence  $\sqrt{n} \text{Exp}_{[\theta_0]}^{-1}(T_n) \in T_{\theta_0} M \cong \mathbb{R}^d$  is bounded in probability as  $n \rightarrow \infty$ . (In symbols, it is  $O_P(1)$ .) Then, for each  $\mathbf{h} \in T_{[\theta_0]} M$ ,

$$\sqrt{n} \Gamma_{\frac{\mathbf{h}}{\sqrt{n}}} \left( \text{Exp}_{[\theta_{n,h}]}^{-1}(T_n) \right) - \sqrt{n} \text{Exp}_{[\theta_0]}^{-1}(T_n) + \mathbf{h} \rightarrow \mathbf{0}$$

in probability as  $n \rightarrow \infty$  when  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  follows  $P_{[\theta_{n,h}]}^n$ .

*Proof.* Fix  $r > 0$  such that the ball  $B_r$  centred at  $\mathbf{0} \in T_{[\theta_0]} M$  with radius  $r$  is compactly contained in  $\mathcal{E}_{[\theta_0]}$  and define  $F : B_r \times B_r \rightarrow T_{[\theta_0]} M$  by setting, for all  $(\mathbf{u}, \mathbf{w}) \in B_r \times B_r$ , that

$$F(\mathbf{u}, \mathbf{w}) = \Gamma_{\mathbf{u}} \left( \text{Exp}_{\text{Exp}_{[\theta_0]}(\mathbf{u})}^{-1} \left( \text{Exp}_{[\theta_0]}(\mathbf{w}) \right) \right).$$

Then  $F$  is a smooth function because the map from  $(\mathbf{u}, \mathbf{z})$  to  $\Gamma_{\mathbf{u}}(\mathbf{z})$  is smooth (as shown with effort in Proposition 10) and the map from  $(\mathbf{u}, \mathbf{w})$  to  $\text{Exp}_{\text{Exp}_{[\theta_0]}(\mathbf{u})}^{-1} \left( \text{Exp}_{[\theta_0]}(\mathbf{w}) \right)$  is smooth. As a consequence, the function

$$R(\mathbf{u}, \mathbf{w}) = F(\mathbf{u}, \mathbf{w}) - (\mathbf{w} - \mathbf{u})$$

satisfies the hypothesis of Lemma 20 and thus

$$\begin{aligned} \left| \sqrt{n} \Gamma_{\frac{\mathbf{h}}{\sqrt{n}}} \left( \text{Exp}_{[\theta_{n,h}]}^{-1}(T_n) \right) - \sqrt{n} \text{Exp}_{[\theta_0]}^{-1}(T_n) + \mathbf{h} \right| &= \sqrt{n} \left| F \left( \frac{\mathbf{h}}{\sqrt{n}}, \text{Exp}_{[\theta_0]}^{-1}(T_n) \right) - \left( \text{Exp}_{[\theta_0]}^{-1}(T_n) - \frac{\mathbf{h}}{\sqrt{n}} \right) \right| \\ &\leq C \left\| \frac{\mathbf{h}}{\sqrt{n}} \right\| \left| \sqrt{n} \text{Exp}_{[\theta_0]}^{-1}(T_n) - \mathbf{h} \right|. \end{aligned}$$

By assumption,  $\sqrt{n} \text{Exp}_{[\theta_0]}^{-1}(T_n)$  is bounded in probability when  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  follows  $P_{[\theta_0]}^n$ . Given that we have proved in Proposition 13 that  $P_{[\theta_0]}^n$  and  $P_{[\theta_{n,h}]}^n$  are mutually contiguous,  $\sqrt{n} \text{Exp}_{[\theta_0]}^{-1}(T_n)$  is also bounded in probability when  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  follows  $P_{[\theta_{n,h}]}^n$ . The proof is therefore complete.  $\square$

**Theorem 22.** *The maximum likelihood estimator is locally regular. With any  $g_{[\theta_0]}$ -orthonormal basis  $\mathbf{e}_1, \dots, \mathbf{e}_d$  of  $T_{[\theta_0]}M$  fixed and the  $d \times d$  matrix  $\mathbf{G}$  defined just as in Theorem 18, we have*

$$\sqrt{n}\Gamma_{\frac{\mathbf{h}}{\sqrt{n}}} \left( \text{Exp}_{[\theta_{n,\mathbf{h}}]}^{-1} \left( \left[ \hat{\theta}_n \right] \right) \right) \rightarrow N_d(\mathbf{0}, \mathbf{G}^{-1})$$

in distribution as  $n \rightarrow \infty$  when  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  follows  $P_{[\theta_{n,\mathbf{h}}]}^n$ .

*Proof.* Put

$$\Lambda_n(\mathbf{h}) = \sum_{i=1}^n \log \frac{p(\mathbf{X}_i; [\theta_{n,\mathbf{h}}])}{p(\mathbf{X}_i; [\theta_0])}.$$

We claim that

$$\begin{bmatrix} \sqrt{n}\text{Exp}_{[\theta_0]} \left( \left[ \hat{\theta}_n \right] \right) \\ \Lambda_n(\mathbf{h}) \end{bmatrix} \rightarrow N_{d+1} \left( \begin{bmatrix} \mathbf{0} \\ -\frac{1}{2}\mathbf{h}^T \mathbf{G} \mathbf{h} \end{bmatrix}, \begin{bmatrix} \mathbf{G}^{-1} & \mathbf{h} \\ \mathbf{h}^T & \mathbf{h}^T \mathbf{G} \mathbf{h} \end{bmatrix} \right) \quad (1.58)$$

in distribution as  $n \rightarrow \infty$  when  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  follows  $P_{\theta_0}^n$ . The asymptotic mean and the diagonal part of the asymptotical covariance matrix are already justified by (1.46) and Theorem 18. Next, by (1.55) and (1.45), we see that the asymptotic covariance term equals to  $\text{Cov}(\mathbf{G}^{-1}\Delta, \Delta^T \mathbf{h})$  with  $\Delta$  being a random variable with distribution  $N_d(\mathbf{0}, \mathbf{G}^{-1})$ , which calculates to  $\mathbf{G}^{-1} \mathbf{G} \mathbf{h} = \mathbf{h}$ .

Next, by Le Cam's third lemma as in Example 6.7 in van der Vaart (1998), (1.58) implies that  $\sqrt{n}\text{Exp}_{[\theta_0]} \left( \left[ \hat{\theta}_n \right] \right)$  converges to  $N_d(\mathbf{h}, \mathbf{G}^{-1})$  in distribution as  $n \rightarrow \infty$  when  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  follows  $P_{[\theta_{n,\mathbf{h}}]}^n$ .

Applying Lemma 21 with  $T_n = \left[ \hat{\theta}_n \right]$  will finish the proof. □

# **Chapter 2**

## **Next Steps**

**2.1 High Dimensional Case**

**2.2 Next Steps in Low Dimensions**

**2.3 Simulations**

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